Introduction

1 What is a pronouncing dictionary?

1.1 Why do we need pronunciation dictionaries?

The simple answer is that the spelling of English is so unreliable as a guide to pronunciation that we need something better. English spelling has evolved over many centuries, but has never been reformed. If it worked as well as the writing systems used for languages such as Finnish or Italian we would know from the spelling exactly how a word should be pronounced, and there would be no point in producing a special dictionary for pronunciations. If at some time in the future the English-speaking world agrees to reform the way English is written, there will probably no longer be a need for our dictionary.

It is very important to remember that in this dictionary we are not trying to tell you how English *ought* to be pronounced; we are presenting how we believe some native speakers of English actually *do* pronounce the words.

This dictionary is designed to provide information on the current pronunciation of approximately 80,000 English words and phrases. For each entry, a British and an American pronunciation is shown. The dictionary provides much essential information that is not usually available in a general dictionary, such as the pronunciation of proper names, the pronunciation of all inflected forms of each word, and a large amount of detail about variant pronunciations. In addition, the electronic versions of the dictionary allow the reader to listen to the British and American pronunciation of a word.

1.2 Can I use the dictionary if I don't know anything about phonetics?

All the information on pronunciation is written using special phonetic symbols based on those of the International Phonetic Association. There is a full explanation of these inside the front cover.

1.3 What is the CD-ROM for?

All the words and all the transcriptions that are in the printed dictionary are also included on the CD-ROM. You can type in any word in the this facility you can listen to a pronunciation of each word (spoken by a British or American voice), then if you wish you can record your own voice and compare it with our version. You can also search for any combination of letters or IPA symbols.

2 The sounds of English

In this section we set out the sounds and the sound structures of the two accents of English (one British and one American) that are the basis of the dictionary. We describe the phonemes and their allophones, then look at syllable structure.

The accent of British English that we describe in this dictionary is called *BBC pronunciation*, since it is the accent most often heard in the speech of newsreaders and announcers on serious channels of the BBC (see Section 3.1 for discussion of the choice of accent); it is also sometimes referred to in this section as *BBC English* or *BBC accent*, often shortened to *BBC*. The corresponding accent of American English is called *General American*, often shortened to *GA*. Many writers have noted that although British and American accents sound very different from each other, the set of sounds (phonemes) used by each is remarkably similar. The differences lie elsewhere. A familiar case is the word 'tomato', where British speakers say /tə'mɑː.təʊ/ and American speakers say /tə'meɪ.ɾoʊ/; British and American speakers both use the phonemes /ɑː/ and /eɪ/ in speaking English, but in the case of 'tomato' they make a different phonemic choice.

To explain one of the most important differences we need to introduce the term *rhotic*. A rhotic accent is one in which the phoneme /r/ is found at the end of syllables as well as at the beginning. In such an accent it is usual for /r/ to be pronounced wherever the letter 'r' occurs in the spelling, so that 'car' is pronounced /kɑːr/ and 'cart' is pronounced /kɑːrt/. Most American accents are rhotic; the British accent described in this dictionary (BBC pronunciation) is *non-rhotic*, though many other accents that can be heard in Britain are rhotic. It is very noticeable in the General American accent that vowels when followed by /r/ are strongly affected ('coloured') by the tongue position required for the production of /r/. The tip of the tongue is curled backwards in a shape that

2.1 Vowels and diphthongs

It is standard practice in phonetics to represent the quality of vowels and diphthongs by placing them in a four-sided figure often known as the *Cardinal Vowel quadrilateral*. This device is used in the vowel descriptions in the following section.

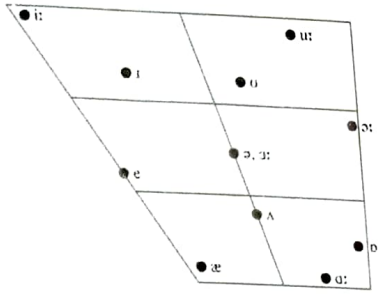


Fig. 1 BBC English pure vowels

2.1.1 British English

British English (BBC pronunciation) is generally described as having short vowels, long vowels, and diphthongs. There are said to be seven short vowels, five long ones, and eight diphthongs. At the end of this section some attention is also given to triphthongs. Each vowel symbol is given a corresponding key-word[1].

- Short vowels:

ɪ	e	æ	ʌ	ɒ	ʊ
kit	dress	trap	strut	lot	foot

The vowel /æ/, classified as a short vowel, is nevertheless considerably longer than the other short vowels before /b, d, g, dʒ, m, n, ŋ/.

In addition, there is the 'schwa' vowel /ə/ which only occurs in weak syllables.

ə	ə
another	

Two additional vowels, /i/ and /u/, are introduced later (Section 3.11).

- Long vowels:

iː	ɑː	ɔː	uː	ɜː
fleece	palm	thought	goose	nurse

- Diphthongs:

eɪ	aɪ	ɔɪ	əʊ
face	price	choice	goat
aʊ	ɪə	eə	ʊə
mouth	near	square	cure

These vowels and diphthongs may be placed in the vowel quadrilateral (based on the Cardinal Vowel diagram) as shown in Figs. 1 – 3. It should be noted that though each vowel (or diphthong starting-point) is marked with a circle, it is misleading to think of this as a precise target; the circle represents the area within which the typical vowel pronunciation falls.

[1] The key-words used here for vowels and diphthongs were devised by Professor J.C. Wells and are widely used throughout the world as a standard set.

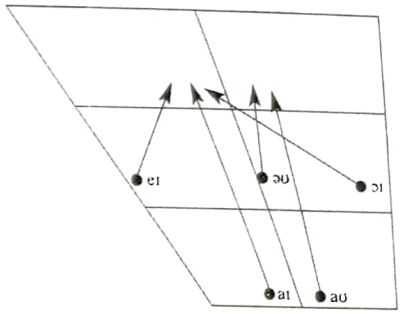


Fig. 2 BBC English closing diphthongs

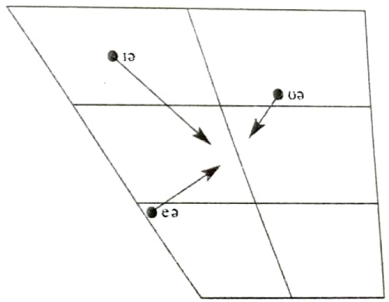


Fig. 3 BBC English centring diphthongs

- Triphthongs:

We need to consider the special case of diphthongs followed by a schwa; examples of these are 'layer' /leɪəʳ/; 'fire' /faɪəʳ/. The full set is as shown below:

/eɪə/	'player'
/aɪə/	'fire'
/ɔɪə/	'loyal'
/əʊə/	'grower'
/aʊə/	'power'

Such sequences are sometimes referred to

, they are not phonemes of English, ...ations of diphthongs with the schwa vowel. However, they present unique ...blems: in BBC pronunciation many of ...ese triphthongs are pronounced with such a small movement in vowel quality that it is difficult for foreign learners to recognize them; for example, the name 'Ireland', which is traditionally transcribed /ˈaɪə.lənd/, frequently has an initial syllable which sounds virtually indistinguishable from /ɑː/, with just a small movement towards /ɪ/ and then towards /ə/ at the end. Some triphthongs are pronounced like single syllables, as in the example just given, while others are more likely to be divided into two syllables (in the list above we have not marked syllable divisions). We usually find the two-syllable pronunciation (i) when the schwa is a separate morpheme (e.g. '-er' in 'buyer' /baɪ.ər/), (ii) when the word is thought to be foreign (this includes many biblical names originating from Hebrew, e.g. 'Messiah' /məˈsaɪ.ə/), and (iii) when a word is not used very frequently, e.g. 'cyanosis' /ˌsaɪ.əˈnəʊ.sɪs/. Where we feel a triphthong may be pronounced either as two syllables or as one, we give one pronunciation with a syllable division and an alternative with a one-syllable pronunciation. In the single-syllable case the middle vowel is printed in italic to indicate that in this reduced pronunciation the middle vowel may be very much reduced or even elided (omitted); for example, 'briar', /ˈbraɪ.ər, braɪər/; where the one-syllable pronunciation seems more likely, that pronunciation is given first: 'fire' is /faər, ˈfaɪ.ər/.

Before an /r/ consonant at the beginning of a following syllable, the distinction between a diphthong and the corresponding triphthong seems to be neutralized, most noticeably in the case of /aɪ/ and /aʊ/ plus /ə/ – it seems that 'Irish', 'irate', 'Maori' may be pronounced as /ˈaə.rɪʃ/, /aəˈreɪt/, /ˈmaʊə.ri/ or as /ˈaɪ.rɪʃ/, /aɪˈreɪt/, /ˈmaʊ.ri/ with no regular distinction made in pronunciation. This is because the glide from /ɪ/ or /ʊ/ to /r/ sounds similar to a schwa. In general, the practice of this dictionary is to transcribe such cases as /aɪə/, /aʊə/; the middle vowel is not printed in italic, to make the transcription less complicated, but it should be understood that the middle vowel in this context may also be considerably weakened.

Speakers of American English are less likely to reduce a triphthong to a single-syllable glide, and this fact is reflected in the transcription given.

A few comments on variation in individual vowels and vowel symbols are needed.

i. The length of vowels and diphthongs is very much reduced when they occur in syllables closed by the voiceless or fortis consonants /p, t, k, tʃ, f, θ, s, ʃ/. Thus /iː/ in 'beat' has only about half the length of /iː/ in 'bead' or 'beet'; similarly /eɪ/ in 'place' is much reduced in length compared with /eɪ/ in 'plays' or 'play'. It is important to remember this in talking about 'long' and 'short' vowels.

ii. The quality of /æ/ is more open than it used to be, and the symbol /a/ might be considered a suitable alternative. We have retained the /æ/ symbol partly because it is phonetically appropriate for the corresponding American vowel.

iii. The vowel /ʌ/ used to be a back vowel in the early years of this dictionary, and the symbol was chosen for this reason. Nowadays it is no longer a back vowel, but a central one. Alternative symbols could be considered in the future.

iv. Among younger speakers, the vowels /uː, ʊ, u/ have moved to a more front quality with less lip rounding, particularly when preceded by /j/ as in 'use'.

v. Among the diphthongs, there seems to be a progressive decline in the use of /ʊə/, with /ɔː/ taking its place (for example, the pronunciation of the word 'poor' as /pɔː/ is increasingly common). The other centring diphthongs also seem to be moving towards a pronunciation more like a long vowel, so that /ɪə/ sounds like [ɪː] and /eə/ sounds like [ɛː] in some speakers.

2.1.2 American English

In American English we do not find the difference between long and short vowels described above, and the vowel system is commonly described as having lax vowels, tense vowels, and diphthongs. Lax vowels, which correspond to British short vowels, are said to be made with less oral tension and do not usually end syllables. The /ɒ/ vowel of British pronunciation is not found in General American. The length of a vowel in American English is principally conditioned by the environment in which it occurs. However, we have retained the length mark on the tense vowels

in order to mark their relationship to the English long vowels.

Vowels preceding /r/ are notably influenced by rhotic colouring. The retroflex vowels /ɜː/ and /ɚ/ are among those features that noticeably distinguish American English pronunciation from BBC pronunciation. All vowels occurring before /r/ within a syllable are likely to become 'r-coloured' to some extent.

- Lax vowels:

ɪ	e	æ	ʌ	ʊ
kit	dress	trap	strut	foot

In addition, there is the schwa vowel and its retroflex counterpart:

about	mother
ə	ɚ

Two other vowels, /i/ and /u/, are introduced below (Section 3.11).

- tense vowels:

iː	ɑː	ɔː	uː	ɜː
fleece	palm	thought	goose	nurse

- Diphthongs:

eɪ	aɪ	ɔɪ	oʊ	aʊ
face	price	choice	goat	mouth

The above vowels and diphthongs may be placed in a vowel quadrilateral as was done for British English (BBC) vowels above. In most cases the positions of the American vowels are so similar to the British ones that the vowel diagrams given (Figs 1 – 3 above) will serve to show the American values, with a few exceptions:

i. There are no centring diphthongs in GA; the equivalents of the BBC vowels shown in Fig. 3 are /ɪr/ (for /ɪə/), /er/ (for eə) and /ʊr/ (for /ʊə/).

ii. There are no contexts in which the /ɜː/ ('nurse') vowel occurs in GA without a following /r/, and we represent this vowel as /ɜː/.

iii. There is no /ɒ/ vowel in GA.

iv. The diphthong in the word 'home' has for many years been represented as /əʊ/ for British pronunciation, but in earlier editions of this and others of Jones' works the symbolization was /oʊ/, indicating a rounded initial vowel. This is still the preferred transcription for the GA diphthong in American books and papers. Therefore, in order to preserve compatibility with other works, we have chosen to use /əʊ/ for BBC English and /oʊ/ for GA. The starting point of /oʊ/ is similar to that of /ɔɪ/, but less back.

v. The GA /æ/ vowel is somewhat closer than BBC /æ/, and seems to be evolving into an even closer vowel. Although it is used in the same words as those with /æ/ in BBC pronunciation, we frequently find /e/ in GA instead. Thus 'marry' is often pronounced /ˈmer.i/, the same pronunciation that can be seen also for 'merry' and 'Mary' (though in some other dialects of US English, notably New York City, there are three different vowels in 'Mary': /ˈmeɪ.ri/, 'merry' /ˈmer.i/, 'marry' /ˈmær.i/). The /æ/ vowel is also found in most of the words which in BBC have /ɑː/ when there is no letter 'r' in the spelling and there is a fricative consonant following the vowel, (e.g. 'pass', 'ask').

vi. The quality of American /ɑː/ is similar to the BBC /ɑː/ vowel; it is used in some of the words which have /ɑː/ in BBC pronunciation when there is no letter 'r' in the spelling (e.g. 'father', 'calm'). It also replaces the BBC short /ɒ/ vowel in many words (e.g. 'hot', 'top', 'bother'): 'bother' rhymes with 'father'.

vii. GA /ɔː/ is more open in quality than BBC /ɔː/. It is used in some words where BBC pronunciation has /ɔː/ (e.g. 'fault', 'wall'), and is also found as a possible pronunciation of words which have the short /ɒ/ in BBC pronunciation, e.g. 'long', 'dog'.

viii. GA /uː/ is similar to BBC /uː/, though slightly more back, but it is also used where BBC has /juː/ after alveolar consonants (e.g. 'new', 'duty').

ix. As explained previously, many GA vowels have 'r-colouring' when they occur before the consonant /r/. Although there are some words where /ɑː/ occurs without this colouring (e.g. 'bra', 'palm'), there are many words like 'car', 'cart' where the /r/ following the /ɑː/ causes the vowel to sound almost like /r/ itself.

2.2 Consonants

2.2.1 British English (BBC pronunciation)

- Plosives:

p	t	k	b	d	g
pin	**t**in	**k**in	**b**in	**d**in	**g**ive

- Affricates:

tʃ	dʒ
chain	**J**ane

- Fricatives:
  f v θ ð
  fine **v**ine **th**ink **th**is
  s z ʃ ʒ h
  seal **z**eal **sh**ip measure **h**ow

- Nasals:
  m n ŋ
  mum **n**un **ng**

- Approximants:
  l r w j
  light **r**ight **w**et **y**et

These consonants (which are common to British and American pronunciation) can be arranged as shown below in the form of a table. The layout of the symbols follows that of the chart of the International Phonetic Alphabet: the horizontal axis corresponds to the place of articulation of the consonant, with the left side of the chart corresponding to the part of the vocal tract furthest from the larynx (i.e. the lips), while the right side is the nearest to the larynx (the place for glottal consonants). From top to bottom are the various types of consonant ('manners of articulation'). Within each box, where there are two consonants which differ only in voicing, they are placed side by side with the voiceless one to the left.

Some explanatory comments about the different consonants are needed.

i. Certain types of consonant have a distinction such as that between /t/ and /d/; this is commonly classed as a distinction between voiceless and voiced consonants, but the distinction is in fact much more complex. Consonants usually classed as voiceless are /p, t, k, f, θ, s, ʃ, tʃ/, with voiced partners /b, d, g, v, ð, z, ʒ, dʒ/. The sound /h/ is also classed as voiceless. Since the presence or absence of voicing is often less important than some other phonetic features, it h been suggested that instead the terms *fo* (equivalent to voiceless) and *lenis* (equivale to voiced) should be used. These terms imp that the main distinguishing factor is t amount of energy used in the articulatio (fortis consonants being made with great energy than lenis). They are not used in th dictionary, since the usefulness of this term nology is uncertain. Some of the characteristi of the two types of consonant are set out belo

ii. /p, t, k/ are typically accompanied b aspiration (i.e. an interval of breath befor the following vowel onset), especially whe initial in a stressed syllable. Thus, 'pin' i distinguished from 'bin' very largely by th aspiration accompanying /p/. However, in th syllable-initial sequences /sp-, st-, sk-/, /p, t, k lack such aspiration. When /l/, /j/, /w/, or /r immediately follow /p, t, k/, they are devoice and are pronounced as fricatives. Anothe characteristic of /p, t, k/ that is not marked in transcriptions is glottalization; when one o these consonants is followed by another con sonant it is now usual to find that a glottal closure precedes the /p/, /t/, or /k/, particu larly if the syllable in which they occur i stressed. Thus the pronunciation of 'captain' 'rightful', 'Yorkshire', which are phonemically /ˈkæp.tɪn/, /ˈraɪt.fᵊl/, /ˈjɔːk.ʃə/, could be shown (using the symbol [ʔ] for glottal closure) a [ˈkæʔp.tɪn], [ˈraɪʔt.fl], [ˈjɔːʔk.ʃə]. This glottaliza tion also occurs in the case of /tʃ/, e.g. 'butcher' /bʊtʃ.əʳ/, [bʊʔtʃ.ə]. When /p/, /t/, /k/, or /tʃ/ occur at the end of a word before a pause they are often glottalized. A related phenomenon can be called 'glottal replacement': younger speakers, in particular, now often pronounce a glottal stop *instead of* /t/ at the end of a stressed syllable preceding an unstressed syl lable (e.g. 'getting better' [geʔ.ŋ beʔ.ə]), but

Table of English Consonants

	Bilabial	*Labio-dental*	*Dental*	*Alveolar*	*Post-alveolar*	*Palatal*	*Velar*	*Glottal*
Plosive	p b			t d			k g	
Affricate					tʃ dʒ			
Fricative		f v	θ ð	s z	ʃ ʒ		(x)	h
Nasal	m			n			ŋ	
Lateral approximant				l				
Approximant	w				r	j		

though some prominent British politicians have adopted this, it is at present not regarded as acceptable BBC pronunciation.

iii. As noted in the section on vowels and diphthongs above, voiceless consonants have a shortening effect on sounds preceding them within a syllable. Thus in the words 'right' and 'ride' (/raɪt/ and /raɪd/) the diphthong is noticeably shorter in the first word than in the second; in the words 'bent' and 'bend' (/bent/ and /bend/), both the vowel /e/ and the nasal consonant /n/ are shorter in the first word. This length difference is not always easy to observe in connected speech.

iv. The consonant /l/ has two different allophones in BBC pronunciation, the so-called 'clear' and 'dark' allophones. The 'clear' one (which has an /iː/-like quality) occurs before vowels, the 'dark' one (which has an /uː/-like quality) before consonants or before a pause. One aspect of the pronunciation of /l/ is becoming much more widespread: in contexts where a 'dark /l/' would be expected, many speakers produce instead a vowel similar to /ʊ/ (for example, in 'hill', 'help'). This is sometimes referred to as '/l/-vocalization'; it has been known to writers on English phonetics for a long time, but has recently become much more widespread.

v. The consonants /ʃ/, /ʒ/, /tʃ/, /dʒ/, /r/ are usually accompanied by lip rounding.

2.2.2 American English

The consonant phonemes of the American English model may be represented by the same scheme used for British English above. Similarly, many of the distinguishing phonetic traits discussed for BBC pronunciation hold for General American as well: initial /p, t, k/ are normally aspirated except when immediately preceded by /s/. Glottalization that precedes the plosives /p, t, k, tʃ/ occurs often in rapid speech. There are, of course, numerous phonetic and phonological differences between British and American English, as there are within regional and social varieties. A few differences are important enough to be mentioned here:

- The 'tapped' medial /t/ following a stressed syllable (as in 'butter', 'water', 'enter') is transcribed as ɾ. This sound is the norm in General American pronunciation; it is usually voiced in conversational speech, but may be devoiced in formalizing environments or in lower-frequency words (which are not shown as tapped in the dictionary). The tap may also occur after /l/ (e.g. 'filter' /ˈfɪl.ɾɚ/), and in this case it is usually voiceless. Since tapping of /t/ after /l/ is not a standard feature of GA, it is not shown in the dictionary.

- Another is phonological, and has already been introduced: the presence (in American English) of postvocalic /r/ (as in 'farmer' /ˈfɑːr.mɚ/), making GA a rhotic accent.

- It should also be noted that the difference between 'clear' and 'dark' /l/ is much less marked in GA than in BBC pronunciation, so that even prevocalic /l/ in American pronunciation sounds dark to English ears. The pronunciation known as '/l/-vocalization', described above in relation to BBC pronunciation, is widely found in contemporary American pronunciation too.

2.3 English syllables

Although vowels and consonants are presented here as the basic building blocks of speech, it is important to remember that each language has its own individual ways of combining those sounds into larger units called syllables. English syllable structure is more complex than that of most of the world's languages. For a description of English syllable structure see Roach (2009), Chapter 8, and more advanced works referred to there. Almost all English syllables have a vowel as their central element, and in the case of an utterance like 'ah' /ɑː/ there is no consonant present. A syllable may have an initial consonant followed by a vowel, e.g. 'bee' /biː/, or a vowel followed by a final consonant, e.g. 'ought' /ɔːt/ (BBC), /ɑːt/ (GA); however, if an English syllable contains one of the 'short' or 'tense' vowels /ɪ, e, æ, ʌ, ɒ, ʊ/ the syllable must end with at least one consonant. Many syllables have an initial and a final consonant, e.g. 'bat' /bæt/. Even with simple syllables like these we must observe certain rules and regularities: for example, the velar nasal /ŋ/ cannot begin a syllable, and the consonant /h/ cannot end one. The rules and regularities become much more complex when we consider syllables containing consonant clusters, that is sequences containing more than one consonant. We find English syllables beginning with two consonants (e.g. 'ski' /skiː/) or three (e.g. 'straw' /strɔː/ (BBC), /strɑː/ (GA)), and syllables ending with two consonants (e.g. 'eats' /iːts/), three (e.g. 'risked' /rɪskt/) or four (e.g. 'sixths'

P

Pronouncing the letter P

→ *See also* **PH**

The consonant letter **p** is most often realized as /p/, e.g.:

pen pen

In addition

p can be silent. There are three combinations in which this can occur: **pn**, **ps** and **pt**.

p is silent in **pn** and **ps** when word initial, e.g.:

pneumatic nju:ˈmæt.ɪk (US) nuːˈmæṭ-
psalm sɑːm

pt can be silent word initially and word finally, e.g.:

pterodactyl ˌter.əʊˈdæk.tɪl (US) ˌter.əˈdæk.ṭəl
receipt rɪˈsiːt

In addition

p can be silent in other instances, e.g.:

corps kɔːʳ (US) kɔːr
cupboard ˈkʌb.əd (US) -əd
raspberry ˈrɑːz.bᵊr.i (US) ˈræz.ber.i

p, P piː -ˈs -z ˌp's and ˈq's
pa pɑː -s -z
PA ˌpiːˈeɪ
Pa. (abbrev. for **Pennsylvania**) ˌpent.sɪlˈveɪ.ni.ə, -sᵊlˈ-, ˈ-njə, (US) -sᵊlˈveɪ.njə, ˈ-ni.ə
pa'anga pɑːˈæŋ.gə, -ə, (US) -ˈɑːŋ.gə, ˈpɑːŋ-
pabulum ˈpæb.jə.ləm, -jʊ-
pace prep ˈpeɪ.si, ˈpɑː.tʃeɪ, -keɪ
pace|n, v peɪs -es -ɪz -ing -ɪŋ -ed -t -er/s -əʳ/z, (US) -ɚ/z ˌput someone ˌthrough their ˈpaces
pacemaker ˈpeɪsˌmeɪ.kəʳ, (US) -kɚ -s -z
pace|man ˈpeɪ.smæn, -smən -men -men
pace-setter ˈpeɪsˌset.əʳ, (US) -ˌseṭ.ɚ -s -z
Pachmann ˈpɑːk.mən, -mɑːn
pachyderm ˈpæk.i.dɜːm, (US) -ə.dɜːm -s -z
pachydermatous ˌpæk.iˈdɜː.mə.təs, (US) -əˈdɜː.mə.ṭəs
pacific, P~ pəˈsɪf.ɪk -ally -ᵊl.i, -li
pacification ˌpæs.ɪ.fɪˈkeɪ.ʃᵊn, ˌ-ə-, -fə-, (US) ˌ-ə- -s -z
pacificatory pəˈsɪf.ɪ.kə.tᵊr.i, pæsˈɪf-, ˌpæs.ɪ.fɪˈkeɪ-, ˌ-ə-; ˈ-ə-, (US) pəˈsɪf.ɪ.kə.tɔːr-
pacificist pəˈsɪf.ɪ.sɪst, pæsˈɪf-, (US) pəˈsɪf- -s -s
pacifism ˈpæs.ɪ.fɪ.zᵊm, ˈ-ə-, (US) ˈ-ə-
pacifist ˈpæs.ɪ.fɪst, ˈ-ə-, (US) ˈ-ə- -s -s
paci|fy ˈpæs.ɪ|.faɪ, ˈ-ə- -fies -faɪz -fying -faɪ.ɪŋ -fied -faɪd -fier/s -faɪ.əʳ/z, (US) -faɪ.ɚ/z
Pacino pəˈtʃiː.nəʊ, (US) -noʊ
pack pæk -s -s -ing -ɪŋ -ed -t -er/s -əʳ/z, (US) -ɚ/z ˈpack ˌice; ˈpacking ˌcase
packag|e ˈpæk.ɪdʒ -es -ɪz -ing -ɪŋ -ed -d ˌpackage ˈholiday; ˈpackage ˌstore
Packard ˈpæk.ɑːd, (US) -ɚd
packed pækt
Packer ˈpæk.əʳ, (US) -ɚ
packet ˈpæk.ɪt -s -s
packhors|e ˈpæk.hɔːs, (US) -hɔːrs -es -ɪz
packing|house ˈpæk.ɪŋ|.haʊs -houses -haʊ.zɪz
pack|man ˈpæk|.mən, -mæn -men -mən, (US) -mən, -men
Pac-man® ˈpæk.mæn
pact pækt -s -s
pactum ˈpæk.təm
pad|y ˈpeɪ.s|i -ier -i.əʳ, (US) -i.ɚ -iest -i.ɪst
pad pæd -s -z -ding -ɪŋ -ded -ɪd ˌpadded ˈcell
Padang ˈpɑː.dɑːŋ, (US) pɑːˈdɑːŋ
Paddington ˈpæd.ɪŋ.tən
paddl|e ˈpæd.ᵊl -es -z -ing -ɪŋ, ˈpæd.lɪŋ -ed -d -er/s -əʳ/z, ˈpæd.ləʳ/z, (US) ˈpæd.ᵊl.ɚ/z, ˈpæd.lɚ/z ˈpaddle ˌsteamer; ˈpaddle ˌwheel
paddleboard ˈpæd.ᵊl.bɔːd, (US) -bɔːrd -s -z
paddling-pool ˈpæd.ᵊl.ɪŋ.puːl, ˈ-lɪŋ- -s -z
paddock, P~ ˈpæd.ək -s -s
paddy, P~ ˈpæd|.i -ies -iz
paddyfield ˈpæd.i.fiːld -s -z
paddywagon ˈpæd.iˌwæg.ən -s -z
Paderewski ˌpæd.əʳˈef.ski, -əʳˈev-, (US) ˌpæd.əˈref-, ˌpɑː.dəˈ-
Padiham ˈpæd.i.əm
Padilla pædˈiː.jə, pəˈdiː-; pəˈdɪl.ə, pædˈɪl-
padlock ˈpæd.lɒk, (US) -lɑːk -s -s -ing -ɪŋ -ed -t
Padraic ˈpɑː.drɪk, ˈpæt.rɪk, (US) ˈpɑː.drɪk, -drɪg
Padraic Colum ˌpɑː.drɪkˈkɒl.əm, ˌpæt.rɪk-, (US) ˌpɑː.drɪkˈkɑː.ləm, -drɪgˈ-
Padraig ˈpɑː.drɪg
padre ˈpɑː.dreɪ, -dri -s -z
padrone pædˈrəʊ.neɪ, pəˈdrəʊ-, -ni, (US) pəˈdroʊ- -s -z
Padstow ˈpæd.stəʊ, (US) -stoʊ
Padu|a ˈpæd.ju|.ə, (US) ˈpædʒ.uː|-, ˌpæd.juˈ|ɑː.grə--er/s -əʳ/z, (US) -ɚ/z
palaeolithic, P~ ˌpæl.i.əʊˈlɪθ.ɪk, ˌpeɪ.li-, (US) ˌpeɪ.li.oʊˈ-, -əˈ-
palaeontological ˌpæl.iˌɒn.təˈlɒdʒ.ɪ.kᵊl, ˌpeɪ.li-, (US) ˌpeɪ.liˌɑːn.təˈlɑː.dʒɪ-
palaeontology ˌpæl.i.ɒnˈtɒl.ə.dʒ|i, ˌpeɪ.li-, (US) ˌpeɪ.li.ɑːnˈtɑː.lə--ist/s -ɪsts
palaeotype ˈpæl.i.əʊ.taɪp, ˈpeɪ.li-, (US) ˈpeɪ.li.oʊ-, -ə-
Palaeozoic ˌpæl.i.əʊˈzəʊ.ɪk, ˌpeɪ.li-, (US) ˌpeɪ.li.oʊˈzoʊ-, -ə-
Palamedes ˌpæl.əˈmiː.diːz
palamino ˌpæl.əˈmiː.nəʊ, (US) -noʊ -s -z
Palamon ˈpæl.ə.mən, -mɒn, (US) -mən, -mɑːn
palanquin ˌpæl.əŋˈkiːn, (US) -ənˈ- -s -z
palatab|le ˈpæl.ə.tə.b|ᵊl, ˈ-ɪ-, (US) -ə.ṭə--ly -li -leness -ᵊl.nəs, -nɪs
palatal ˈpæl.ə.tᵊl, (US) -ṭᵊl -s -z
palatalization ˌ-isa- ˌpæl.ə.tᵊl.aɪˈzeɪ.ʃᵊn, pəˌlæt.ᵊl-, -ɪˈ-, (US) ˌpæl.ə.ṭᵊl.əˈ-s -z
palataliz|e ˌ-is|e ˈpæl.ə.tᵊl.aɪz, pəˈlæt.ᵊl-, (US) ˈpæl.ə.ṭə.laɪz -es -ɪz -ing -ɪŋ -ed -d
palate ˈpæl.ət, -ɪt, (US) -ət -s -s
palatial pəˈleɪ.ʃᵊl, (US) -ˈʃᵊl-ly -i
palatinate, P~ pəˈlæt.ɪ.nət, -ᵊn.ət, -ɪt, (US) -ᵊn.eɪt, -ɪt -s -s
palatine, P~ ˈpæl.ə.taɪn, (US) -taɪn, -tɪn -s -z
palatogram ˈpæl.ə.təʊ.græm, pəˈlæt.əʊ-, (US) ˈpæl.ə.toʊ-, -ṭə--s -z
palatography ˌpæl.əˈtɒg.rə.fi, (US) -ˈtɑː.grə-
palav|er pəˈlɑː.v|əʳ, (US) -ˈlæv|.ɚ, -ˈlɑː.v|ɚ--ers -əz, (US) -ɚz-ering -ᵊr.ɪŋ-ered -əd, (US) -ɚd
palazzo pəˈlæt.səʊ, -sə; ˈ-ˈlæd.zəʊ, -zə, (US) pəˈlɑːt.soʊ, pɑː- pa|lazzo ˈpants
pal|e peɪl -er -əʳ, (US) -ɚ -est -ɪst, -əst -ely -li -eness -nəs, -nɪs -es -z -ing -ɪŋ -ed -d ˌpale ˈale; beˌyond the ˈpale
palefac|e ˈpeɪl.feɪs -es -ɪz
paleo- ˈpæl.i.əʊ, ˈpeɪ.li.əʊ, (US) ˈpeɪ.li.oʊ, -ə
paleobotany ˌpæl.i.əʊˈbɒt.ᵊn.i, ˌpeɪ.li-, (US) ˌpeɪ.li.oʊˈbɑː.tᵊn-
Paleocene ˈpæl.i.əʊ.siːn, ˈpeɪ.li-, (US) ˈpeɪ.li.oʊ-, -ə-
paleographic ˌpæl.i.əʊˈgræf.ɪk, ˌpeɪ.li-, (US) ˌpeɪ.li.oʊˈ-, -əˈ-
paleography ˌpæl.i.əʊˈɒg.rə.f|i, ˌpeɪ.li-, (US) ˌpeɪ.liˈɑː.grə--er/s -əʳ/z, (US) -ɚ/z
paleolithic, P~ ˌpæl.i.əʊˈlɪθ.ɪk, ˌpeɪ.li-, (US) ˌpeɪ.li.oʊˈ-, -əˈ-
paleontological ˌpæl.iˌɒn.təˈlɒdʒ.ɪ.kᵊl, ˌpeɪ.li-, (US) ˌpeɪ.liˌɑːn.təˈlɑː.dʒɪ-
paleontology ˌpæl.i.ɒnˈtɒl.ə.dʒ|i, ˌpeɪ.li-, (US) ˌpeɪ.li.ɑːnˈtɑː.lə--ist/s -ɪsts
paleotype ˈpæl.i.əʊ.taɪp, ˈpeɪ.li-, (US) ˈpeɪ.li.oʊ-, -ə-
Palermo pəˈlɜː.məʊ, -ˈleə-, (US) -ˈler.moʊ
Palestine ˈpæl.ə.staɪn, ˈ-ɪ-, (US) -ə.staɪn
Palestinian ˌpæl.əˈstɪn.i.ən, -ɪˈ-, (US) -əˈstɪn--s -z
Palestrina ˌpæl.əˈstriː.nə, -ɪˈ-, (US) -əˈstriː-
palette ˈpæl.ət, -ɪt, -et, (US) -ɪt -s -s ˌpalette ˌknife
Paley ˈpeɪ.li
palfrey, P~ ˈpɔːl.fri, ˈpɒl-, (US) ˈpɔːl-, ˈpɑːl- -s -z
Palgrave ˈpɔːl.greɪv, ˈpæl-, (US) ˈpæl-, ˈpɔːl-
Pali ˈpɑː.li
palimony ˈpæl.ɪ.mə.ni, ˈ-ə-, (US) -ə.moʊ-
palimpsest ˈpæl.ɪmp.sest, -əmp-, (US) -ɪmp--s -s
Palin ˈpeɪ.lɪn
palindrome ˈpæl.ɪn.drəʊm, -ən-, (US) -ɪn.droʊm-s -z
palindromic ˌpæl.ɪnˈdrɒm.ɪk, -ənˈ-, (US) -ɪnˈdrɑː.mɪk, - droʊ-
paling ˈpeɪ.lɪŋ-s -z
palingenesis ˌpæl.ɪnˈdʒen.ə.sɪs, ˈ-ɪ-, (US) -ə-
palinode ˈpæl.ɪ.nəʊd, ˈ-ə-, (US) -ə.noʊd-s -z
palisad|e, P~ ˌpæl.ɪˈseɪd, -ə-, (US) -ə-, ˈ----es -z-ing -ɪŋ-ed -ɪd
palish ˈpeɪ.lɪʃ
Paik pɔːlk, pɒlk, (US) pɔːlk, pɔːk
pall pɔːl-s -z-ing -ɪŋ-ed -d
palladi|an, P~ pəˈleɪ.di|.ən-um/s -əm/z
Pallas ˈpæl.əs, -æs
pall-bearer ˈpɔːl|beə.rəʳ, (US) -ˌber.ɚ, ˈpɑːl--s -z
pallet ˈpæl.ɪt, -ət, (US) -ɪt-s -s
palliass|e ˈpæl.i.æs, ˌ--ˈ-, (US) pælˈjæs, ˈ---es -ɪz
palli|ate ˈpæl.i|.eɪt-ates -eɪts -ating -eɪ.tɪŋ, (US) -eɪ.ṭɪŋ-ated -eɪ.tɪd, (US) -eɪ.ṭɪd
palliation ˌpæl.iˈeɪ.ʃᵊn
palliative ˈpæl.i.ə.tɪv, (US) -ṭɪv-s -z
pallid ˈpæl.ɪd-est -ɪst, -əst-ly -li -ness -nəs, -nɪs
Palliser ˈpæl.ɪ.səʳ, (US) -sɚ
palli|um ˈpæl.i|.əm-ums -əmz-a -ə
pall-mall ˌpælˈmæl, (US) ˌpælˈmæl, ˌpelˈmel, ˌpɑːlˈmɑːl
Pall Mall ˌpælˈmæl, ˌpelˈmel, (US) ˌpælˈmæl, ˌpelˈmel, ˌpɔːlˈmɔːl *stress shift:* ˌPall Mall ˈClub
pallor ˈpæl.əʳ, (US) -ɚ
pally ˈpæl.i
palm pɑːm-s -z-ing -ɪŋ-ed -d ˌPalm ˈBeach; ˌPalm ˈSprings; ˌPalm ˈSunday; ˌgrease one's ˈpalm
Palma ˈpæl.mə, ˈpɑː-, ˈpɑːl-, (US) ˈpɑːl.mɑː
palmar ˈpæl.məʳ, -mɑːʳ, (US) -mɚ, ˈpɑːl-
palmat|e ˈpæl.meɪ|t, ˈpɑː-, -mɪ|t, (US)

Note: Usually pronounced pəˈhʌŋ/ in Malaysia.

Pronouncing the letters PH

The consonant digraph **ph** is usually pronounced as /f/, e.g.:

photo	ˈfəʊ.təʊ (US) ˈfoʊ.ṭoʊ
alphabet	ˈæl.fə.bet

However, the realization /v/ can occur in some words, e.g.:

nephew	ˈnef.juː, ˈnev-
Stephen	ˈstiː.vᵊn

In addition

A much less common realization of the consonant digraph **ph** is /p/, e.g.:

shepherd	ˈʃep.əd (US) -ɚd

petty bourgeoisie ˌpet.iˌbɔː.ʒwɑːˈzi, -ˌbʊə-, (US) ˌpeṭ.iˌbʊr-

Petula pɪˈtʃuː.lə, pə-, -ˈtjuː-, petˈuː-, (US) petˈjuː-, pəˈtuː-, pəˈtjuː-

petulan|ce ˈpetʃ.ə.lənt|s, ˈpet.jʊ-, ˈpet.jə-, ˈpetʃ.ʊ-, (US) ˈpetʃ.ə- **-cy** -si

petulant ˈpetʃ.ə.lənt, ˈpet.jʊ-, ˈpet.jə-, ˈpetʃ.ʊ-, (US) ˈpetʃ.ə- **-ly** -li

petunia pɪˈtʃuː.ni.ə, pə-, -ˈtjuː-, ˈ-njə, (US) pəˈtuː.njə, -ˈtjuː-, ˈ-ni.ə **-s** -z

Petworth ˈpet.wəθ, -wɜːθ, (US) -wɚθ, -wɝθ

Peugeot® ˈpɜː.ʒəʊ, (US) pɜːˈʒoʊ, puː-, pjuː- **-s** -z

Pevensey ˈpev.ᵊn.zi

Peveril ˈpev.ᵊr.ɪl

Pevsner ˈpevz.nəʳ, (US) -nɚ

phallicism ˈfæl.ɪ.sɪ.zᵊm

phallocentric ˌfæl.əʊˈsen.trɪk, (US) -oʊ-, -əˈ- *stress shift:* ˌphallocentric ˈculture

phall|us ˈfæl|.əs **-uses** -ə.sɪz **-i** -aɪ

phanerogam ˈfæn.ᵊr.əʊ.gæm, fəˈner-, (US) ˈfæn.ə.roʊ-, -ɚ.ə-; (US) fəˈner.oʊ-, ˈ-ə- **-s** -z

phanerogamic ˌfæn.ᵊr.əʊˈgæm.ɪk, fəˌner-, (US) ˌfæn.ə.roʊˈ-, -ɚ.əˈ-, fəˌner.oʊˈ-, -əˈ-

phanerogamous ˌfæn.ᵊrˈɒg.ə.məs, (US) -əˈrɑː.gə-

phantasm ˈfæn.tæz.ᵊm **-s** -z

phantasmagoria ˌfæn.tæz.məˈgɔː.ri.ə, -təz-, -ˈgɒr.i-, (US) -ˈgɔːr.i-

phantasmagoric ˌfæn.tæz.məˈgɒr.ɪk, -təz-, (US)

pharynx ˈfær.ɪŋks **-es** -ɪz **pharynges** fərˈɪn.dʒiːz, fəˈrɪn-, (US) fə-

phas|e feɪz **-es** -ɪz **-ing** -ɪŋ **-ed** -d

phas|is ˈfeɪ.s|ɪs, (US) -s|ɪs, -z|ɪs **-es** -iːz

phatic ˈfæt.ɪk, (US) ˈfæṭ-

PhD ˌpiː.eɪtʃˈdiː **-s** -z

pheasant ˈfez.ᵊnt **-s** -s

Phebe ˈfiː.bi

Phèdre ˈfed.rə

Phelps felps

phenacetin fɪˈnæs.ɪ.tɪn, fə-, fenˈæs-, ˈ-ə-, (US) fɪˈnæs.ə-, -tən

Phenic|ia fɪˈnɪʃ|.ə, fə-, fiː-, -ˈniː.ʃ|ə, -i|.ə, (US) fəˈnɪʃ|.ə, -ˈniː.ʃ|ə **-ian/s** -ᵊn/z

pheno- ˈfiː.nəʊ, (US) ˈfiː.noʊ, ˈ-nə

phenobarbitone ˌfiː.nəʊˈbɑː.bɪ.təʊn, -bə-, (US) -noʊˈbɑːr.bɪ.toʊn, -nəˈ-

phenol ˈfiː.nɒl, (US) -noʊl, -nɔːl, -nɑːl

U

Pronouncing the letter U

→ *See also* **UE, UI, UOU, UY**

The vowel letter **u** has several strong pronunciations linked to spelling. 'Short' pronunciations include /ʌ/ and /ʊ/, and 'long' pronunciations include /uː/ and /juː/. In 'short' pronunciations pronounced /ʌ/, the **u** is generally followed by a consonant letter which ends the word, or a double consonant before another vowel. Words containing /ʊ/ which end with a consonant sound often have two consonant letters finally, a notable exception being *put* /pʊt/, e.g.:

tub	tʌb
tubby	ˈtʌb.i
bull	bʊl
bully	ˈbʊl.i

The 'long' pronunciations usually mean the **u** is followed by a single consonant letter and then a vowel, e.g.:

tube	tʃuːb (US) tuːb
tubing	ˈtʃuː.bɪŋ (US) ˈtuː-
supervise	ˈsuː.pə.vaɪz (US) -pɚ-

In word-initial position, the 'long' pronunciation is almost always pronounced /juː/, e.g.:

unique	juːˈniːk
useful	ˈjuːs.fəl

However, there are exceptions to these guidelines, e.g.:

study	ˈstʌd.i
sugar	ˈʃʊg.ər (US) -ɚ
truth	truːθ

When **u** is followed by **r**, the strong pronunciation is one of several possibilities: /jʊə, jɔː/ (US) /jʊr/, /ʊə, ɔː/ (US) /ʊr/, /ɜː/ (US) /ɜːˈ/, or /ʌ/ (US) /ɜːˈ/, e.g.:

cure	kjʊəʳ, kjɔːʳ (US) kjʊr
plural	ˈplʊə.rəl, ˈplɔː- (US) ˈplʊr.əl
burn	bɜːn (US) bɜːˈn
hurry	ˈhʌr.i (US) ˈhɜːˈ.i

An exceptional case is /e/ in bury /ˈber.i/.

In weak syllables

The vowel letter **u** is realized as one of /jə/, /jʊ/, /ə/, or /ʊ/ in weak syllables, e.g.:

failure	ˈfeɪ.ljəʳ (US) ˈfeɪl.jɚ
accurate	ˈæk.jə.rət, -jʊ- (US) -jɚ.ət, -jʊ.rət
status	ˈsteɪ.təs (US) ˈstæ.ṭəs
July	dʒʊˈlaɪ

It may also result in a syllabic consonant, e.g.:

hopeful	ˈhəʊp.fəl, -fʊl (US) ˈhoʊp-

u, U juː -'s -z
UAE juː.eɪˈiː
UB40 ˌjuː.biːˈfɔː.ti, (US) -ˈfɔːr.ṭi -s -z
Ubbelohde ˈʌb.əl.əʊd, (US) -oʊd
U-bend ˈjuː.bend -s -z
über- ˈuː.bəʳ, (US) -bɚ
Übermensch ˈuː.bə.mentʃ, (US) -bɚ- -en -ən
ubiquitarian, U~ juː.ˌbɪk.wɪˈteə.ri.ən, ˌjuː.bɪ.kwɪˈ-, (US) juː.ˌbɪk.wəˈter.i- -s -z
ubiquitous juːˈbɪk.wɪ.təs, (US) -wə.ṭəs -ly -li -ness -nəs, -nɪs
ubiquity juːˈbɪk.wə.ti, -wɪ-, (US) -wə.ṭi
U-boat ˈjuː.bəʊt, (US) -boʊt -s -s
Ubu Roi ˌuː.buːˈrwɑː
UCAS, Ucas ˈjuː.kæs
UCATT ˈʌk.ət, ˈjuː.kæt
UCCA ˈʌk.ə
Uccello uːˈtʃel.əʊ, (US) -oʊ
Uckfield ˈʌk.fiːld
UCLA ˌjuː.siː.elˈeɪ
UDA ˌjuː.diːˈeɪ
Udall ˈjuː.dəl, (US) -dɔːl, -dɑːl, -dəl
udder ˈʌd.əʳ, (US) -ɚ -s -z
Uddin ˈʌd.ɪn
Uddingston ˈʌd.ɪŋ.stən
UDI ˌjuː.diːˈaɪ
Udolpho uːˈdɒl.fəʊ, juː-, (US) -ˈdɑːl.foʊ
UDR ˌjuː.diːˈɑːʳ, (US) -ˈɑːr
UEFA, Uefa juːˈeɪ.fə, -ˈiː-
Uffizi juːˈfɪt.si, uː-, -ˈfiːt-
UFO ˌjuː.efˈəʊ, (US) -ˈoʊ -(')s -z
ufology juːˈfɒl.ə.dʒ|i, (US) -ˈfɑː.lə- -ist/s -ɪst/s
Uganda juːˈgæn.d|ə, jʊ-, (US) juː-, uː- -an/s -ən/z
Ugaritic ˌuː.gəʳˈrɪt.ɪk, ˌjuː-, (US) -gəˈrɪṭ-
ugh ʌx, ɜːh, ʌg

> Note: Used to indicate disgust. Many other pronunciations are possible, including [ɯːx], [ɯh], [uɸ].

ugli ˈʌg.li -s -z ˈugli ˌfruit
uglification ˌʌg.lɪ.fɪˈkeɪ.ʃən, -lə-, -fəˈ-
uglify ˈʌg.lɪ|.faɪ, ˈ-lə- -fies -faɪz -fying -faɪ.ɪŋ -fied -faɪd
ugly ˈʌg.l|i -ier -i.əʳ, (US) -i.ɚ -iest -i.ɪst, -i.əst -iness -ɪ.nəs, -ɪ.nɪs ˌugly ˈduckling
Ugrian ˈuː.gri.ən, ˈjuː- -s -z
Ugric ˈuː.grɪk, ˈjuː-
UHF ˌjuː.eɪtʃˈef
uh-huh ʌˈhʌ, ˈʌ.hʌ
uhlan ˈuː.lɑːn, ˈjuː-, -lən; uːˈlɑːn, juː- -s -z
Uhland ˈuː.lənd, -lænd, -lɑːnd, (US) -lɑːnt, -lɑːnd
uh-oh ˈʌʔ.əʊ, (US) -oʊ
UHT ˌjuː.eɪtʃˈtiː *stress shift, see compound:* ˌUHT ˈmilk
Uhu® ˈjuː.huː, ˈuː-
uh-uh ˈʌʔˌʌʔ, (US) ˈÃʔˌÃ

> Note: Pronounced with falling pitch, this has the meaning 'no'.

uhuru, U~ uːˈhuː.ruː, -ˈhʊə-
Uig ˈuː.ɪg, ˈjuː-
Uighur ˈwiː.gʊəʳ, -gəʳ, (US) -gʊr, -gɚ -s -z
Uist ˈjuː.ɪst
uitlander, U~ ˈeɪtˌlæn.dəʳ, ˈɔɪt-, (US) ˈɔɪtˌlæn.dɚ, ˈeɪt-, ˈaɪt- -s -z
UK ˌjuːˈkeɪ *stress shift:* ˌUK ˈcitizen
ukase juːˈkeɪz, -ˈkeɪs, (US) juːˈkeɪz, -ˈkeɪs, ˈ-- -es -ɪz
uke juːk -s -s
ukelele ˌjuː.kəˈleɪ.li, (US) -kəˈleɪ- -s -z
UKIP ˈjuː.kɪp
Ukraine juːˈkreɪn, (US) -ˈkreɪn, ˈjuː.kreɪn -ian/s -iən/z
ukulele ˌjuː.kəˈleɪ.li, (US) -kəˈleɪ- -s -z
ulan ˈuː.lɑːn, ˈjuː-, -lən; uːˈlɑːn, juː- -s -z
Ulan Bator uːˌlɑːnˈbɑː.tɔːʳ, (US) ˌuː.lɑːnˈbɑː.tɔːr
Ulan-Ude uːˌlɑːn.uːˈdeɪ, (US) uːˌlɑːn.uːˈdeɪ, uːˈlɑːn.uːˌdeɪ

Glossary

abbreviations

The pronunciation of individual abbreviations is not predictable and must be treated on a word-by-word basis.

Examples

Some good examples of abbreviations which are spoken in full are to be found in titles used for people, e.g.:

Mr /ˈmɪs.tər/ (US) /-tɚ/
Mrs /ˈmɪs.ɪz/
Dr /ˈdɒk.tər/ (US) /ˈdɑːk.tɚ/
Esq /ɪˈskwarər/ (US) /ˈes.kwaɪ.ɚ/

However, in some cases, an abbreviation may be pronounced the way it is written, e.g.:

Bros. (*Brothers*) /ˈbrʌð.əz/, /brɒs/ (US) /ˈbrʌð.ɚz/
des res (*desirable residence*) /ˌdezˈrez/

Some words or phrases are abbreviated to initial letters. In this case, the item may be pronounced as the initial letters, or in full. This is not the same as an ACRONYM, in which the letters are pronounced as a word (see, for example, *NATO*), e.g.:

MS (*multiple sclerosis; manuscript*) /ˌemˈes/
NBA (*National Basketball Association*) /ˌenˈbiːˈeɪ/
UCLA (*University of California Los Angeles*) /ˌjuː.siː.elˈeɪ/

Abbreviations derived from Latin words and phrases are common, but do not all follow the same pattern, some being pronounced in full, some as letters, and some as they are spelt, e.g.:

e.g. (*exempli gratia*) /ˌiːˈdʒiː/
et al (*et alia*) /etˈæl/ (US) /-ˈɑːl/
etc. (*et cetera*) /ɪtˈset.ər.ə/ (US) /-ˈseṭ.ɚ-/
i.e. (*id est*) /ˌaɪˈiː/

accent

Accent may refer to prominence given to a syllable (see STRESS), or to a particular way of pronouncing. This entry concentrates on the latter definition.

Examples for English

Speakers may share the same grammar and vocabulary, but pronounce what they say with different accents. In Britain, for example, there are accents such as Scots, Cockney (in London), and Scouse (in Liverpool), and in the United States the New York accent differs considerably from that commonly heard in Texas. There are also different world standard English accents, such as British, American, Australian, Indian, or Singaporean.

Major differences between British and US English include the vowel in words such as *glass*, the use of a flapped /t/ in words like *butter*, and the fact that US English is RHOTIC while British English is non-rhotic, that is, an *r* in the spelling is always pronounced in US English, but only where a vowel follows in British English, e.g.:

glass /glɑːs/ (US) /glæs/
butter /ˈbʌt.ə/ (US) /ˈbʌṭ.ɚ/
car /kɑːˈ/ (US) /kɑːr/

More information on the accents chosen to represent British and American English is given in the Introduction.

acronyms

Acronyms are words or phrases written in an abbreviated form, usually with their initial letters, and pronounced as if they were words, usually with the most obvious pronunciation. However, it is best to treat them on a word-by-word basis, since there are exceptions.

Examples

Many organizations, particularly those to do with government, charities, and trade unions, use acronyms. Phrases can also be reduced to acronym form. Acronyms are usually written using capital letters, although there are exceptions. Where there is a letter *E* at the end of an acronym, it is usually pronounced /iː/, e.g.:

- NATFHE (*National Association of Teachers in Further and Higher Education*) /ˈnæt.fiː/
- NATO (*North Atlantic Treaty Organization*) /ˈneɪ.təʊ/ (US) /-ṭoʊ/
- snafu (*situation normal, all fouled up*) /snæfˈuː/
- Tardis (*time and relative dimensions in space*) /ˈtɑː.dɪs/ (US) /ˈtɑːr-/

Some words or phrases are abbreviated to initial letters but do not become acronyms. See ABBREVIATIONS.

affricate

A type of consonant consisting of a PLOSIVE followed by a FRICATIVE with the same place of ARTICULATION.

Examples for English

Examples are the /tʃ/ and /dʒ/ sounds (sometimes symbolized /č/ and /ǰ/ by American writers) at the beginning and end of the words *church* and *judge*, where the first of these sounds is voiceless and the second voiced (see VOICING), e.g.:

church /tʃɜːtʃ/ (US) /tʃɜːtʃ/
judge /dʒʌdʒ/

It is often difficult to decide whether any particular combination of a plosive plus a fricative should be classed as a single affricate sound or as two separate sounds, and the question depends on whether these are to be regarded as separate PHONEMES or not. It is usual to regard /tʃ/ and /dʒ/ as affricate phonemes in English; /ts dz tr dr/ also occur in English but are not usually regarded as affricate phonemes, but are treated as clusters. The two phrases *why choose* and *white shoes* are said to show the difference between the /tʃ/ affricate (in the first example) and separate /t/ and /ʃ/ (in the second), e.g.:

why choose /*h*waɪ tʃuːz/
white shoes /*h*waɪt ʃuːz/

allophone

A variant of a phoneme.

Examples for English

Central to the concept of the PHONEME is the idea that it may be pronounced in many different ways. In English we take it for granted that the /r/ sounds in *ray* and *tray* are 'the same sound' (i.e. the same phoneme), but in reality the two sounds are very different – the /r/ in *ray* is voiced and non-fricative, while the /r/ sound in *tray* is voiceless and FRICATIVE. In phonemic transcription we use the same symbol /r/ for both (the slant brackets indicate that phonemic symbols are being used), but we know that the allophones of /r/ include the voiced non-fricative sound and the voiceless fricative one. Using the square brackets that indicate PHONETIC (allophonic) symbols, the former is [ɹ] and the latter [ɹ̠], e.g.:

ray /reɪ/ [ɹeɪ]
tray /treɪ/ [tr̥eɪ]

In theory, a phoneme can have an infinite number of allophones, but in practice for descriptive purposes we tend to concentrate on the ones that occur most regularly and recognizably.

alveolar

Alveolar sounds are made with a place of articulation behind the upper front teeth, against the hard, bony ridge called the alveolar ridge; the skin covering it is corrugated with transverse wrinkles.

Examples for English

The tongue comes into contact with the alveolar ridge in some of the consonants of English and many other languages; sounds such as [t], [d], [s], [z], [n], [l] are consonants with alveolar place of articulation. Some examples for English follow, e.g.:

tip /tɪp/
zip /zɪp/
dip /dɪp/
nip /nɪp/
sip /sɪp/
lip /lɪp/

Although /r/ is described as alveolar or post-alveolar in British English, in US English it is in fact nearer to RETROFLEX, e.g.:

rip /rɪp/ [ɹɪʔp] (US) [ɹɪʔp]

approximant

A phonetic term of comparatively recent origin, used to denote a consonant which makes very little obstruction to the airflow.

Examples for English

Traditionally approximants have been divided into two groups. Sounds in the first group are known as 'semivowels' such as the /w/ and /j/, which are very similar to 'close vowels' such as [u] and [i] but are produced as a rapid glide, e.g.:

wet /wet/
yet /jet/

'Liquids' are sounds which have an identifiable constriction of the airflow but not one that is sufficiently obstructive to produce fricative noise, compression, or the diversion of airflow through another part of the vocal tract as in nasals. This category includes laterals such as /l/ and non-fricative /r/ (phonetically [ɹ] in British English and [ɹ̠] in US English), e.g.:

lead /liːd/
read /riːd/

Approximants therefore are never fricative and never contain interruptions to the flow of air.

articulation

The movement of the vocal organs to produce speech sounds. The vocal organs are often referred to as 'articulators', and these include the tongue, the lips, the hard palate, the soft palate, the teeth, the pharynx, and the larynx. In classifying CONSONANTS, phoneticians note the 'place of articulation' (the point in the vocal tract where the obstruction to the airflow is made) and the 'manner of articulation' (the type of obstruction made by the articulators). Thus the sound [s] is classified as ALVEOLAR (because the place of articulation is at the alveolar ridge, just behind the upper front teeth) and FRICATIVE (because the obstruction is one which allows air to escape with difficulty, creating a hissing noise).

aspiration

Noise made when the constriction of a plosive consonant is released and air is allowed to escape relatively freely.

Examples for English

English /p t k/ at the beginning of a syllable are aspirated in most accents. In words like *pea, tea, key*, there is a silent period during which the compressed air is prevented from escaping by the articulatory closure; this is followed by a sound similar to /h/ before the VOICING of the vowel begins, the result of the vocal folds being widely parted at the time of the articulatory release. Aspiration is an important factor in whether we perceive a sound to be /p t k/ or /b d g/ in syllable-initial position.

/p t k/ are aspirated at the beginning of a syllable containing a full vowel, e.g.:

pin /pɪn/ [pʰɪn]
tick /tɪk/ [tʰɪʔk]
kin /kɪn/ [kʰɪn]
appease /əˈpiːz/ [əˈpʰiːz̩]
attain /əˈteɪn/ [əˈtʰeɪn]
accord /əˈkɔːd/ [əˈkʰɔːd̪] (US) /əˈkɔːrd/ [əˈkʰɔːɹd̪]

When followed by /l r w/ or /j/ in initial consonant clusters, the release of the PLOSIVE gives the following sound a voiceless quality, e.g.:

play /pleɪ/ [pl̥eɪ]
tree /triː/ [tr̥iː]
queue /kjuː/ [kj̊uː]
twice /twaɪs/ [tw̥aɪs]

It is noticeable that when /p t k/ are preceded by /s/ at the beginning of a syllable they are not aspirated. This makes them very similar to /b d g/, and is an example of NEUTRALIZATION – the distinctive difference

between /p t k/ and /b d g/ is lost when there is a preceding /s/, e.g.:

spin /spɪn/ [spɪn] (compare pɪn [pʰɪn])
stick /stɪk/ [stɪʔk] (compare tɪk [tʰɪʔk])
skin /skɪn/ [skɪn] (compare kɪn [kʰɪn])

Word final /p t k/ may be aspirated or not, for stylistic reasons.

assimilation

Assimilation is what happens to a sound when it is influenced by one of its neighbours; essentially it becomes more similar to a neighbour.

Examples for English

Assimilation is said to be 'progressive' when a sound influences a following sound, or 'regressive' when a sound influences one which precedes it. The most familiar case of regressive assimilation in English is that of ALVEOLAR consonants (e.g. /t d s z n/) which are followed by non-alveolar consonants: assimilation results in a change of place of articulation from alveolar to a different place. For example, the word *this* has the sound /s/ at the end if it is pronounced on its own, but when followed by post-alveolar /ʃ/ in a word such as *shop* it often changes in rapid speech (through assimilation) to /ʃ/, giving the pronunciation /ðɪʃʃɒp/. The following examples occur especially in British English:

batman /'bæt.mæn/ → ['bæp.mæn]
fruitcake /'frut.keɪk/ → ['fruk.keɪk]
handbag /'hænd.bæg/ → ['hæmb.bæg]

Progressive assimilation is exemplified by the behaviour of the 's' plural ending in English, which is pronounced with a voiced /z/ after a voiced consonant but with a voiceless /s/ after a voiceless consonant, e.g.:

dog /dɒg/ US /dɑːg/ + plural → /dɒgz/ US /dɑːgz/
cat /kæt/ + plural → /kæts/

bilabial

Bilabial articulations involve both of the lips.

Examples for English

In English, /p b m/ are examples of bilabial sounds. These are all made with a complete closure of the lips. /w/ is also sometimes referred to as bilabial, but, as it has tongue movement towards the velum in addition to lip rounding, it is more accurately described as 'labial-velar'. E.g.:

pan /pæn/
ban /bæn/
man /mæn/

The plosives /p/ and /b/ are one of the pairs which are said to be distinguished by being FORTIS and LENIS respectively, rather than voiceless and voiced.

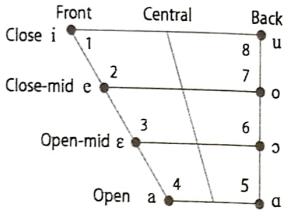
cardinal vowel

One of the vowels of the standard classification system used in phonetics.

Description

Phoneticians have always needed some way of classifying vowels which is independent of the vowel system of a particular language. With most consonants it is quite easy to observe how their articulation is organized, and to specify the place and manner of the constriction formed; vowels, however, are much less easy to observe.

Early in the 20th century, the English phonetician Daniel Jones worked out a set of 'cardinal vowels' that students learning phonetics could be taught to make and which would serve as reference points that other vowels could be related to. The vowels are located on the four-sided figure shown below:



The cardinal vowel figure is used to specify the qualities of the English vowels and diphthongs in the Introduction to this dictionary.

The cardinal vowel diagram is used both for rounded and unrounded vowels, and Jones proposed that there should be a primary set of cardinal vowels and a secondary set. The primary includes the front unrounded vowels [i e ɛ a], the back unrounded vowel [ɑ], and the rounded back vowels [ɔ o u], while the secondary set comprises the front rounded vowels [y ø œ ɶ], the back rounded [ɒ], and the back unrounded [ʌ ɤ ɯ].

Phonetic 'ear-training' makes much use of the cardinal vowel system, and students can learn to identify and discriminate a very large number of different vowels in relation to the cardinal vowels.

clear l

A type of LATERAL sound in which the air escapes past the sides of the tongue, found normally only before vowels. Usually contrasted with DARK L.

Examples for English

In the case of an alveolar lateral (e.g. English /l/) the blade of the tongue (the part further back than the tip) is in contact with the alveolar ridge, but the rest of the tongue is free to take up different shapes. One possibility is for the front of the tongue (the part behind the blade) to be raised in the same shape as that for a close front vowel [i]. This gives the /l/ an [i]-like sound, and the result is a 'clear l'. It is found in BBC pronunciation only before vowels, but in some other accents, notably Irish and Welsh ones, it is found in all positions. However, the variant most often found in all positions is 'DARK L', e.g.:

pill /pɪl/ [pʰɪɫ]

cluster

Two or more CONSONANT PHONEMES in sequence, with no vowel sound between them.

Examples for English

English can allow up to three consonants in a cluster syllable initially in the ONSET, and four finally in the CODA. For example, the word *stray* /streɪ/ begins with three consonants, and *sixths* /sɪksθs/ ends with four.

Some types of grammatical information are shown by adding certain consonants at the ends of words in English, giving rise to consonant clusters, e.g.:

likes /laɪks/
lives /lɪvz/
liked /laɪkt/
lived /lɪvd/
Mark's /mɑːks/ US /mɑːrks/
John's /dʒɒnz/ US /dʒɑːnz/
cats /kæts/
dogs /dɒgz/ US /dɑːgz/

coalescence

A kind of ASSIMILATION in which a fusion of neighbouring sounds takes place during rapid or CONNECTED SPEECH.

Examples for English

The most frequently observed situation in which coalescence occurs in English is when ALVEOLAR consonants /t d s z/ are followed by /j/, e.g.:

won't you /wəʊnt ju/ US /woʊnt ju/ → /'wəʊntʃu/ US /'woʊntʃu/
would you /wʊd ju/ → /'wʊdʒu/

In the case of coalescence with /s z/, there will usually be extra length to the resulting fricative sounds, e.g.:

miss you /mɪs ju/ → /'mɪʃʃu/
lose you /luːz ju/ → /'luːʒʒu/

A common very much reduced example of coalescence is *do you* being pronounced as /dʒə/.

coarticulation

The influence of phonetic context on the articulation of speech sounds.

Examples for English

Phonetics studies coarticulation as a way of finding out how the brain controls the production of speech. When we speak, many muscles are active at the same time and sometimes the brain tries to make them do things that they are not capable of. For example, in the word *mum* the vowel phoneme is one that is normally pronounced with the soft palate raised to prevent the escape of air through the nose, while the two /m/ phonemes must have the soft palate lowered. The soft palate cannot be raised very quickly, so the vowel is likely to be pronounced with the soft palate still lowered, giving a nasalized quality to the vowel, e.g.:

mum /mʌm/ [mɐ̃m]

Another example is the lip rounding of a consonant in the environment of rounded vowels: in the phrase 'you too', the /t/ occurs between two rounded vowels, and there is not enough time in normal speech for the lips to move from rounded to unrounded and back again in a few hundredths of a second; consequently the /t/ is pronounced with lip rounding, e.g.:

you too /juː tuː/ [jʉːtʷʉː]

Coarticulation is a phenomenon closely related to ASSIMILATION. The major difference is that assimilation is used as a name for the process whereby one sound becomes like another neighbouring sound, while coarticulation, though it refers to a similar process, is concerned with articulatory explanations for why the assimilation occurs, and considers cases where the changes may occur over a number of segments.

coda

The end of a syllable, which is said to be made up of an ONSET, a peak, and a coda. The peak and the coda constitute the RHYME (or RIME) of the syllable.

Examples for English

English allows up to four consonants to occur in the coda, so the total number of possible codas in English is very large – several hundred in fact, e.g.:

sick /sɪk/
six /sɪks/
sixth /sɪksθ/
sixths /sɪksθs/

The central part of a syllable is almost always a vowel, and if the syllable contains nothing after the vowel it is said to have no coda ('zero coda'), e.g.:

bough /baʊ/
buy /baɪ/

compounds

Compounds are words made up of two other words which can exist independently in English. They are written in a variety of ways: closed, e.g. *armchair, sunflower*; hyphenated, e.g. *front-runner, she-devil*; open, e.g. *side salad, bank manager*. Sometimes it is possible for the same compounds to be written in different ways, e.g. *shoulder blade* or *shoulderblade*.

There are some rules for compound word stress in English, but they are not completely reliable. In addition, British and US English differ. The rules which follow are therefore guidelines, but exceptions exist.

Examples

Words that combine two nouns usually have primary stress on the first element, e.g.:

coffee pot /'kɒf.i.pɒt/ US /'kɑː.fi.pɑːt/
suitcase /'suː.keɪs/

In US English it is more frequently the case that compound nouns have first-element stress, and this tendency is appearing among some British English speakers. For example, *ice cream* can have first- or second-element stress in British English, but tends to be stressed on the first element in US English.

Other types of compound are usually stressed on the

second element. These include words ending in *-ed* which are used as adjectives, words with a number as a first element, and words functioning as adverbs or verbs, e.g.:

kind-hearted /ˌkaɪndˈhɑːɹtɪd/ (US) /-ˈhɑːɹtɪd/
three-piece /ˌθriːˈpiːs/
southeast /ˌsaʊθˈiːst/

These words are subject to stress shift when a stressed syllable follows closely (e.g. *a ˌkind-hearted ˈman*).

connected speech phenomena

The processes which result in words being pronounced differently from their dictionary form when they occur in close contact with other words.

Examples for English

In natural speech we rarely leave gaps between words, and we can observe many processes that result in differences between isolated words and the same words occurring in connected speech: examples are ASSIMILATION and ELISION, e.g.:

assimilation

one boy /wʌn bɔɪ/ → /wʌm bɔɪ/

elision

last time /lɑːst taɪm/ (US) /læst taɪm/ → /lɑːs taɪm/
(US) /læs taɪm/

The study of connected speech also involves looking at the process of VOWEL REDUCTION in weak syllables (see also WEAK FORM), at RHYTHM and at prosodic phenomena such as INTONATION and STRESS.

consonant

A speech sound which obstructs the flow of air though the vocal tract or a letter of the alphabet representing such a vowel.

Examples for English

There are many types of consonant, but what all have in common is that they obstruct the flow of air through the vocal tract. Some do this a lot, some not very much: those which make the maximum obstruction (i.e. PLOSIVES, which form a complete stoppage of the airstream) are the most consonantal, e.g.:

bat /bæt/
keep /kiːp/

NASAL consonants result in complete stoppage of the oral cavity but are less obstructive than plosives since air is allowed to escape through the nose, e.g.:

man /mæn/
name /neɪm/

FRICATIVES make a considerable obstruction to the flow of air, but not a total closure, e.g.:

sheaf /ʃiːf/
south /saʊθ/

The class of sounds called APPROXIMANTS comprises sounds which make very little obstruction to the flow of air. The post-alveolar approximant that is the usual pronunciation of /r/ in BBC pronunciation involves no contact between the tongue and the palate, e.g.:

red /red/ [ɹɛd]
arrive /əˈraɪv/ [əˈɹaɪv]

LATERAL approximants obstruct the flow of air only in the centre of the mouth, not at the sides, so obstruction is slight, e.g.:

lull /lʌl/

Other sounds classed as approximants make so little obstruction to the flow of air that they could almost be thought to be vowels if they were in a different context. These are sometimes referred to as 'semi-vowels', e.g.:

you /juː/
woo /wuː/

Finally, AFFRICATES begin as plosives and are released as fricatives, with no intervening gap, e.g.:

church /tʃɜːtʃ/ (US) /tʃɝːtʃ/
judge /dʒʌdʒ/

The above explanation is based on phonetic criteria. An alternative approach is to look at the *phonological* characteristics of consonants: for example, consonants are typically found at the beginning and end of syllables while vowels are typically found in the middle.

dark l

A type of LATERAL sound, recognizably different from CLEAR L.

Examples for English

If, when pronouncing the sound /l/, the back of the tongue is raised as for an [u] vowel, the quality is [u]-like and 'dark'; this effect is even more noticeable if the lips are rounded at the same time. This sound is typically found when /l/ occurs before a consonant or before a pause, e.g.:

help /help/ [hɛɫp]
hill /hɪl/ [hɪɫ]

In several accents of English, particularly those close to London, the dark l has given way to a [w] sound, so that *help* and *hill* might be transcribed /hewp/ and /hɪw/; this process is known as 'l vocalization'.

In other languages

The process of l vocalization took place in Polish some time ago, and the sound represented in Polish writing by the letter ł is almost always pronounced as [w], though foreigners usually try to pronounce it as [l].

dental

A sound in which there is contact between the tongue and the front teeth.

Examples for English

In English, the dentals usually referred to are the FRICATIVES /θ/ and /ð/, of which /θ/ is voiceless and /ð/ is voiced. In a careful production of these sounds, the tongue tip may be protruded between the upper and lower teeth; the sounds are sometimes referred to as 'interdental' for this reason, e.g.:

thigh /θaɪ/
ether /ˈiː.θəʳ/ (US) /-θɚ/
breath /breθ/
thy /ðaɪ/
either /ˈaɪ.ðəʳ/ (US) /ˈiː.ðɚ/
breathe /briː.ð/

devoicing

A process affecting a sound which we would normally expect to be voiced but which is pronounced without VOICING in a particular context.

Examples for English

In English, the /l/ in *blade* /bleɪd/ [bleɪd] is usually voiced, but in *played* /pleɪd/ [pl̥eɪd] the /l/ is usually voiceless because of the preceding voiceless plosive. Note also that the /d/ at the end of the word in each case is devoiced if it is followed by a voiceless consonant or a pause.

The notion of devoicing leads to a rather confusing use of phonetic symbols in cases where there are separate symbols for voiced and voiceless pairs of sounds: a devoiced /d/ can be symbolized by adding a diacritic that indicates lack of voice – [d̥], but one is then left in doubt as to what the difference is between this sound and [t]. The usual reason for doing this is to leave the symbol looking like the phoneme it represents.

diphthong

A sound in which there is a glide from one vowel quality to another.

Examples for English

BBC English and US English contain a large number of diphthongs: in both accents, there are three ending in [ɪ] (/eɪ aɪ ɔɪ/), sometimes referred to as 'front-closing', and two ending in [ʊ] (/aʊ əʊ/), sometimes referred to as 'back-closing'. In US English, the preferred transcription of the BBC vowel /əʊ/ is /oʊ/, which indicates a rounded initial vowel.

BBC English also has three diphthongs ending in [ə] (/ɪə eə ʊə/), which are sometimes known as 'centring'. The /ʊə/ diphthong is now less commonly used than /ɔː/. These diphthongs usually appear in US English with an /r/ final (/ɪr er ʊr/), as words containing them generally end with an *r* in the spelling, e.g.:

bay /beɪ/
buy /baɪ/
boy /bɔɪ/
go /gəʊ/ (US) /goʊ/
cow /kaʊ/
pier /pɪə/ (US) /pɪr/
pear /peə/ (US) /per/
poor /pʊə/ (more often /pɔː/) (US) /pʊr/

Opinions differ as to whether diphthongs should be treated as phonemes in their own right, or as combinations of two phonemes.

elision

The omission of sounds which are normally present if words are pronounced slowly and clearly but appear not to be pronounced when the same words are produced in a rapid, colloquial style, or when the words occur in a different context. These missing sounds are said to have been 'elided'. See also CONNECTED SPEECH PHENOMENA.

Examples for English

It is easy to find examples of elision, but very difficult to state rules that govern which sounds may be elided and which may not. Elision of vowels in English usually happens when a short, unstressed vowel occurs between voiceless consonants, e.g. in the first syllable of *perhaps*, *potato*, the second syllable of *bicycle*, or the third syllable of *philosophy*. Elision also occurs when a weak vowel occurs between a PLOSIVE or FRICATIVE consonant and a consonant such as a NASAL or a LATERAL: this process leads to SYLLABIC CONSONANTS, e.g.:

sudden /ˈsʌd.ən/ → /ˈsʌd.n̩/
awful /ˈɔː.fʊl/ (US) /ˈɑː/ → /ˈɔː.fl̩/ (US) /ˈɑː-/

Elision of consonants in English happens most commonly when a speaker 'simplifies' a complex consonant cluster, e.g.:

acts /ækts/ → /æks/
twelfth night /ˌtwelfθˈnaɪt/ → /ˌtwelθˈnaɪt/ or /ˌtwelfˈnaɪt/

In *twelfth night* above, it seems much less likely that any of the other consonants could be left out: the /l/ and the /n/ seem to be unelidable.

It is very important to note that sounds do not simply 'disappear' like a light being switched off. A transcription such as /æks/ for *acts* implies that the /t/ phoneme has dropped out altogether, but detailed examination of speech shows that such effects are more gradual: in slow speech the /t/ may be fully pronounced, with an audible transition from the preceding /k/ and to the following /s/, while in a more rapid style it may be articulated but not given any audible realization, and in very rapid speech it may be observable, if at all, only as a rather early movement of the tongue blade towards the /s/ position. Much more research in this area is needed (not only on English) for us to understand what processes are involved when speech is 'reduced' in rapid articulation.

flap

A type of consonant sound that is closely similar to the TAP. It is usually voiced, and is produced by slightly curling back the tip of the tongue, then throwing it forward and allowing it to strike the alveolar ridge as it descends. The phonetic symbol for the sound is [ɾ]

Examples in English

This sound, although occurring in some accents of English, is not a PHONEME, and is uncommon in British English. It appears, however, in US English, where it is sometimes heard in words like *party* and *birdie*, where the /r/ consonant causes retroflexion of the tongue

and the stress pattern favours a flap-like articulation. In this dictionary we represent the former as /ɾ/, e.g.:

party /ˈpɑː.ti/ (US) /ˈpɑːr.ɾi/ [ˈpʰɑːɹ̩.ɾi]
birdie /ˈbɜː.di/ (US) /ˈbɜː.di/ [ˈb̪ɜː.ɾi]

In other languages

A flap is most commonly heard in languages which have other RETROFLEX consonants, such as languages of the Indian subcontinent; it is also heard in the English of native speakers of such languages, often as a realization of /r/.

fortis

Fortis sounds are said to be made with a relatively high degree of effort.

Examples for English

It is claimed that in some languages (including English) there are pairs of consonants whose members can be distinguished from each other in terms of whether they are 'strong' (fortis) or 'weak' (LENIS). These terms refer to the amount of energy used in their production, and are similar to the terms TENSE and LAX more usually used in relation to vowels. It is argued that English /b d g v ð z ʒ/ often have little or no voicing in normal speech, and it is therefore a misnomer to call them voiced. Since they seem to be more weakly articulated than /p t k f θ s ʃ/ it would be appropriate to use the term lenis instead, e.g.:

pin /pɪn/ [pʰɪ̈n]
bin /bɪn/ [b̥ɪ̈n]
fine /faɪn/ [faɪ̈n]
vine /vaɪn/ [v̩aɪ̈n]

French words and phrases

French has provided a substantial proportion of English vocabulary since the Norman Conquest in 1066, with new words being incorporated all the time. The longest-established and the most commonly used words are usually the most completely anglicized, and the most noticeable changes from the French original are found where English phonology does not easily accommodate them. Often there are many possible anglicized pronunciations: see for example the many listed for *restaurant*.

Examples

Word-final /r/: this is not usually pronounced in British English unless followed by a vowel in a following word. However, it is pronounced in US English, e.g.:

savoir faire /ˌsæv.wɑːˈfeəʳ/ (US) /-wɑːrˈfer, -wɑːˈ-/

Nasalized vowels: although some English speakers who have a good command of French may attempt to produce the French nasalized vowels /ɛ̃/, /ɑ̃/, /ɔ̃/, and even occasionally /œ̃/, these are pronounced by many people with a following /ŋ/.

Croissant is given the pronunciation /ˈkwæs.ɑ̃ːŋ/ for British English, indicating that the vowel in the last syllable will be pronounced with a nasalized vowel but possibly also a velar nasal consonant. In US English, however, a velar nasal is not usually produced following a nasalized vowel, and *croissant* is pronounced /kwɑːˈsɑ̃ː/. (It should be noted that in both cases other pronunciations exist.) This example also shows the simplification of the initial consonant cluster /krw/, which occurs in French but not in English.

Word-final stress: many French words and names with final stress are modified by British English speakers to have the stress at the beginning of the word. In US English, this occurs in some words, but not in others, e.g.:

ballet /ˈbæl.eɪ/ (US) /bælˈeɪ, ˈ- -/
Paris /ˈpær.ɪs/ (US) /ˈper-, ˈpær-/
restaurant /ˈres.tʳɒ.ɔ̃ːŋ/ (US) /-tə.rɑːnt/

The phoneme /ʒ/: while this phoneme occurs frequently in the middle of words, it is rare at the beginning and end. Many such cases are in words of French origin. English speakers sometimes substitute /dʒ/ for this sound, e.g.:

gigolo /ˈdʒɪg.ə.ləʊ, ˈʒɪg-/ (US) /-loʊ/
garage /ˈgær.ɑːʒ, -ɪdʒ, -ɑːdʒ/ (US) /gəˈrɑːʒ, -ˈrɑːdʒ/

fricative

A type of consonant made by forcing air through a narrow gap so that a hissing noise is generated. This may be accompanied by VOICING, in which case the sound is a voiced fricative, such as [z], or it may be voiceless, such as [s].

Examples for English

British and US English have nine fricative phonemes: /f θ s ʃ h/ (voiceless) and /v ð z ʒ/ (voiced).

All except /h/ are permitted to occur in all positions in English, but /ʒ/ as in 'measure' /ˈmeʒə/ is of rather low frequency compared to the other eight sounds. /h/ may not end a syllable. The quality and intensity of fricative sounds varies greatly, but all are acoustically composed of energy at relatively high frequency – an indication of this is that much of the fricative sound is too high to be transmitted over a phone (which usually cuts out the highest and lowest frequencies in order to reduce the cost), giving rise to the confusions that often arise over sets of words like English *fin*, *thin*, *sin*, and *shin*. In order for the sound quality to be produced accurately, the size and direction of the jet of air has to be very precisely controlled. A distinction is sometimes made between 'sibilant' or 'strident' fricatives (such as [s] and [ʃ]), which are strong and clearly audible, and others which are weak and less audible (such as [θ] and [f]).

glottalization

The addition of a glottal stop before a consonant.

Examples for English

Adding a glottal stop before certain consonants has the effect of making the preceding vowel somewhat shorter. In English this usually happens before a voiceless PLOSIVE or AFFRICATE consonant if there is not a vowel immediately following (e.g. in *captive*, *catkin*, *arctic*; a similar case is that of /tʃ/ when following a stressed vowel or when syllable final, as in *butcher*).

This addition of a glottal stop is sometimes called 'glottal reinforcement', e.g.:

back /bæk/ [b̥æʔk]
captive /ˈkæp.tɪv/ [ˈkʰæʔp̚.tɪv]
catkin /ˈkæt.kɪn/ [ˈkʰæʔk̚.kɪn]
arctic /ˈɑːk.tɪk/ [ˈɑːʔk̚.tɪk] (US) /ˈɑːrk.tɪk/ [ˈɑːɹʔk̚.tɪk]
butcher /ˈbʊtʃ.əʳ/ [ˈbʊʔ.tʃ̬ ə] (US) /ˈbʊtʃ.ɚ/ [ˈbʊʔ.tʃ̬ɚ]

This feature of English is an important one for perception. As the difference in voicing between /p t k tʃ/ and /b d g dʒ/ in syllable-final position is negligible, it is the length of the vowel rather than the voicing of the final consonant which contributes strongly to a native speaker's decision of whether a speaker has produced e.g. *back* or *bag*.

glottal stop

A consonant made by closure of the vocal folds. The phonetic symbol for a glottal stop is [ʔ].

Examples for English

In some British accents, a glottal stop can actually replace the voiceless alveolar plosive [t] as the realization of the /t/ phoneme when it follows a stressed vowel, e.g.:

getting better /ˌget.ɪŋˈbet.əʳ/ → [ˌge̞ʔ.ɪ̃ŋˈbe̞ʔ.ə]

This type of pronunciation is found in many urban accents, notably London (Cockney), Leeds, Glasgow, Edinburgh, and others, and is increasingly accepted among educated young people. Sometimes a glottal stop is pronounced in front of a /p/, /t/, or /k/ if there is not a vowel immediately following (see GLOTTALIZATION).

In a true glottal stop there is complete obstruction to the passage of air, and the result is a period of silence. In casual speech it often happens that a speaker aims to produce a complete glottal stop but instead makes a low-pitched creak-like sound.

homographs

When two lexical items have the same form in spelling, these are known as homographs. Homographs can be pronounced the same as or different from each other.

Examples

Here are some examples of homographs with the same pronunciation:

- bank /bæŋk/ (*e.g. financial institution; area of ground by a river*)
- well /wel/ (*e.g. source of water; healthy; adv.*)

Other homographs differ in their pronunciation, but usually only in the vowel sound used, e.g.:

- bow
  — (*e.g. knot; device for shooting arrows*) /bəʊ/ (US) /boʊ/
  — (*e.g. incline the head or trunk; submit*) /baʊ/
- dove
  — (*pigeon*) /dʌv/
  — (*past tense of* dive) /dəʊv/ (US) /doʊv/

Homographs also appear in noun/verb, noun/adjective, and verb/adjective pairs. The difference in pronunciation indicates which part of speech is being used, e.g.:

insert (*n.*) /ˈɪn.sɜːt/ (US) /-sɜːt/
(*v.*) /ɪnˈsɜːt/ (US) /-ˈsɜːt/
deliberate (*adj.*) /dɪˈlɪb.ʳ.ət/
(*v.*) /dɪˈlɪb.ʳ.eɪt/ (US) /-ə.reɪt/
arithmetic (*n.*) /əˈrɪθ.mə.tɪk/
(*adj.*) /ˌær.ɪθˈmet.ɪk/ (US) /-er.ɪθˈmeṭ-/

Pronunciation tip

In most cases it is necessary to check the pronunciation of a word individually, as the correct realization is not obvious from the form of the word. However, some rules are available. For example, the pronunciation of word pairs ending with *-ate* is predictable depending on the part of speech. In two-syllable words where the stress moves to show the difference between a noun and a verb (e.g. *insert*, *export*), the noun is almost always stressed on the first syllable and the verb on the second.

intonation

The use of the pitch of the voice to convey linguistic information. The word is used with two rather different meanings. In a restricted sense, it is the variations in the pitch of a speaker's voice used to convey or alter meaning. In a broader and more popular sense, intonation is equivalent to 'prosody', where variations in such things as voice quality, tempo, and loudness are included.

Examples for English

Intonation is said to convey emotions and attitudes. Other linguistic functions have also been claimed: interesting relationships exist in English between intonation and grammar, for example. In a few extreme cases a perceived difference in grammatical meaning may depend on the pitch movement, e.g.:

- she didn't go because of her ˇtimetable (meaning 'she did go, but it was not because of her timetable')
- she didn't go because of her ˋtimetable (meaning 'she didn't go, the reason being her timetable').

Other 'meanings' of intonation include things like the difference between statement and question, e.g.:

- it was ˋcold (meaning 'it was cold')
- it was ˊcold (meaning 'was it cold?')

the contrast between 'open' and 'closed' lists, e.g.:

- would you like ˊwine, ˊsherry or ˇbeer ('open', implying other things are also on offer)
- would you like ˊwine, ˊsherry or ˋbeer ('closed', implying no further choices are available)

and the indication of whether a relative clause is restrictive or non-restrictive, e.g.:

- the students who were ˇnervous ˋfailed (restrictive relative clause: only students who were nervous failed)

▸ the ˋstudents, who were ˋnervous, ˋfailed (non-restrictive relative clause: all students were nervous and all failed)

Another approach to intonation is to concentrate on its role in conversational discourse: this involves such aspects as indicating whether the particular thing being said constitutes new information or old, the regulation of turn-taking in conversation, the establishment of dominance, and the elicitation of co-operative responses. As with the signalling of attitudes, it seems that though analysts concentrate on pitch movements, there are many other prosodic factors being used to create these effects.

labiodental

A consonant articulated with contact between the lips and the teeth.

Examples for English

By far the most common type of labiodental articulation is one where the inner surface of the lower lip touches the upper front teeth, as in the fricatives [f] (voiceless) and [v] (voiced); these two occur in English, e.g.:

fine /fam/
safe /serf/
vine /vam/
save /serv/

The fricative noise made by /f/ and /v/ is very weak. In final position, as /f/ is FORTIS and /v/ is LENIS, 'pre-fortis clipping' of the vowel occurs. This has the effect of shortening the vowel in *safe*, making it much shorter than the one in *save*.

larynx

The larynx is located in the throat and its main biological function is to act as a valve that can stop air entering or escaping from the lungs and also (usually) prevents food and other solids from entering the lungs. It consists of a rigid framework or box made of cartilage and, inside, the vocal folds, which are two small lumps of muscular tissue like a very small pair of lips with the division between them (the 'glottis') running from front to back of the throat. There is a complex set of muscles inside the larynx that can open and close the vocal folds as well as changing their length and tension.

In speech the larynx has many important functions including the following:

i) the distinction between voiced and voiceless sounds

ii) the control of pitch

iii) the production of the glottal fricative [h] and the glottal stop [ʔ]

iv) producing variation in voice quality.

lateral

A lateral consonant is one where there is obstruction to the passage of air in the centre (mid-line) of the air passage and the air flows to the side of the

Examples for English

In English the /l/ phoneme is lateral both in its 'clear' and its 'dark' allophones (see CLEAR L and DARK L): the blade of the tongue is in contact with the alveolar ridge as for /t d n/ but the sides of the tongue are lowered to allow the passage of air, e.g.:

lip /lɪp/
pill /pɪl/

When an alveolar plosive precedes a lateral consonant in English it is usual for it to have a 'lateral release'. This means that to go from /t/ or /d/ to /l/ we simply lower the sides of the tongue to release the compressed air, rather than lowering and then raising the tongue blade. A syllabic /l/ is the usual result of this in word final position (see SYLLABIC CONSONANT), e.g.:

bottle /ˈbɒt.l/
puddle /ˈpʌd.l/

Most laterals are produced with the air passage to both sides of the obstruction (they are 'bilateral'), but sometimes we find air passing to one side only ('unilateral').

In other languages

Other lateral consonants are found in other languages: the Welsh *ll* sound is a voiceless lateral fricative [ɬ], and Xhosa and Zulu have a voiced lateral fricative [ɮ]. Several Southern African languages have lateral clicks (where the plosive occlusion is released laterally) and at least one language (of Papua New Guinea) has a contrast between alveolar and velar lateral.

Latin words and phrases

Words, names, and phrases from Latin have entered the English language at many different times. Some words and names are in relatively common use and have been completely anglicized, while others are used in particular types of discourse which to some extent determine their pronunciation. The pronunciation used by academic scholars of Latin has tended to be based on a reconstruction of what was supposed to be the pronunciation in Roman times. Ecclesiastical Latin, previously used in the Roman Catholic church and closely similar to the pronunciation of Italian, has largely disappeared, but phrases such as *Gloria in excelsis* or *Humanae Vitae* are still heard. Legal Latin is also now much less widely used than it used to be, but some phrases survive, such as *habeas corpus, ultra vires* (usual pronunciation /ˌheɪ.bi.əsˈkɔː.pəs, US /-ˈkɔːr-/ and /ˌʌl.trəˈvaɪə.riːz, ˌʊl.traːˈvɪə.reɪz/ US /ˌʌl.trəˈvaɪ.riːz/). Most of the Latin words, phrases, and names in this dictionary are ones that have been fully anglicized, but there is no set of rules to determine exactly how this is done.

Examples

Words, e.g.:

accidia /ækˈsɪd.i.ə/
flamen /ˈfleɪ.men/
vale /ˈvɑː.leɪ, ˈveɪ.li, ˈvæl.eɪ/ US /ˈveɪ.li, ˈvɑː.leɪ/

Names, e.g.:

Aeneas /iːˈniː.əs, ɪˈniː-, -æs/

Phrases, e.g.:

ad hoc /ˌæd ˈhɒk, -ˈhaʊk/ US /-ˈhɑːk, -ˈhoʊk/
ex voto /ˌeks ˈvəʊ.təʊ/ US /-ˈvoʊ.toʊ/
flagrante delicto /flə.ɡræn.teɪ.dɪˈlɪk.təʊ/, /flæɡ.ræn-, -tɪ-, -dəˈ-, -deɪˈ-/ US /flə.ɡræn.ti.diːˈlɪk.toʊ/

lax

A lax sound is produced with relatively little articulatory energy. Since there is no established standard for measuring articulatory energy, this concept only has meaning if it is used relative to some other sounds that are felt to be articulated with a comparatively greater amount of energy (i.e. TENSE).

Examples for English

It is mainly American phonologists who use the terms lax and tense in describing English vowels; the short vowels /ɪ e æ ʌ ʊ ə/ are classed as lax, while what are referred to in our description of BBC pronunciation as the long vowels and the diphthongs are tense. The terms can also be used of consonants as equivalent to FORTIS (tense) and LENIS (lax), though this is not commonly done in present-day descriptions.

length

A term used in phonetics to refer to a subjective impression of how much time a sound takes; it is distinct from physically measurable 'duration'. Usually, however, the term is used as synonymous with duration.

Examples for English

Length is important in many ways in speech: in English and most other languages, stressed syllables tend to be longer than unstressed (see RHYTHM, STRESS and WEAK FORM). Some languages have phonemic differences between long and short sounds, and BBC English is claimed by some writers to be of this type, contrasting short vowels /ɪ e æ ʌ ɒ ʊ ə/ with long vowels /iː ɜː ɑː ɔː uː/ (though other, equally valid analyses have been put forward). However, the context in which these sounds occur must be taken into account. For example, the vowel /iː/ is said to be longer than /ɪ/ as well as having a different quality, but the vowel in *beat* is unlikely to be longer than the vowel in *bid* as the phonetic environment in *beat* causes the vowel to be shorter.

In other languages

When languages have long/short consonant differences, as does Arabic, for example, it is usual to treat the long consonants as geminate; it is odd that this is not done equally regularly in the case of vowels. Perhaps the most interesting example of length differences comes from Estonian, which has traditionally been said to have a threeway distinction between short, long, and extra-long consonants and vowels.

lenis

A lenis sound is weakly articulated (the word comes from Latin, where it means 'smooth, gentle'). The opposite term is FORTIS.

Examples for English

In general, the term lenis is used of voiced consonants (which are supposed to be less strongly articulated than their corresponding voiceless ones), and is resorted to for languages such as German, Russian, and English where voiced phonemes like /b d ɡ/ are not always voiced. (See the entry at FORTIS for examples.) However, it is claimed that the language which most clearly shows a distinction between fortis and lenis consonants is Korean.

liaison

The linking or joining together of sounds.

Examples for English

In English the best-known case of liaison is the 'linking r': there are many words in English (e.g. *car, here, tyre*) which in a RHOTIC accent such as US English or Scots would be pronounced with a final /r/, but which in BBC pronunciation end in a vowel when they are pronounced before a pause or before a consonant. When they are followed by a vowel, British English speakers pronounce /r/ at the end, e.g.:

the car stopped /ðə kɑː stɒpt/ US /ðə kɑːr stɑːpt/
the car is blue /ðə kɑːr ɪz bluː/ US /ðə kɑːr ɪz bluː/

In BBC English there is also 'intrusive r' – an /r/ inserted between two vowels at word boundaries where there is none in the spelling. This does not occur after close vowels (/iː uː/), or diphthongs which end with a close element (/eɪ aɪ ɔɪ aʊ əʊ/), e.g.:

China and Japan /tʃaɪnə r ən dʒə pæn/ US /tʃaɪnə ən dʒə pæn/
law and order /lɔː r ən ɔː.də/ US /lɑː ən ɔːr.dɚ/

It has been argued that this elision has the purpose of avoiding bringing two vowels into contact with each other, but since many languages do run vowels together with no apparent difficulty for speakers, the argument should be treated with caution.

Another aspect of liaison in English is the movement of a single consonant at the end of an unstressed word to the beginning of the next if that is strongly stressed. A well known example in British English is *none at all*, where the /t/ of *at* becomes initial (and therefore strongly aspirated) in the final syllable for many speakers.

monophthong

A single vowel. The term is used only in contrast with the word DIPHTHONG, which originally meant a 'double sound'.

Examples for English

British English has 12 vowel monophthongs /ɪ e æ ʌ ɒ ʊ ə iː ɑː ɔː ɜː uː/, and US English has 11, or 12 if r-coloured SCHWA /ɚ/ is taken into account. In British English these are traditionally divided into short and long, with a length mark [ː] used to show that there is a difference in length as well as vowel

quality. This convention is extended in this dictionary to US English vowels. Long vowels are permitted to appear in a stressed syllable without a CODA, whereas short vowels are not. It should be noted that the schwa vowel, /ə/, never appears in stressed syllables and has a different distribution from the other short vowels.

names of people and places

It can be difficult to work out the pronunciation of some English words, and something like between ten to thirty percent of words in any text will have irregular spellings. Proper nouns for people and places can have really unexpected pronunciations. Here, we look at a few of the most interesting ones.

Examples

Family names are well known for having interesting realizations. This can be because some letters are not pronounced, but, in some cases, the way a word is written looks almost entirely different from the pronunciation, e.g.:

Cholmondeley	/ˈtʃʌm.li/
Colquhoun	/kəˈhuːn/
Dalziel	/diː.el; ˈdæl.ziːl/
Featherstonehawe	/ˈfeð.ə.stən.hɔː; ˈfæn.ʃɔː/ Ⓤⓢ /ˈfeð.ɚ.stən.hɑː; ˈfæn.ʃɑː/
Quesnel	/ˈkeɪ.nᵊl/

It is a similar situation for place names, e.g.:

Alnwick	/ˈæn.ɪk/
Cirencester	/ˈsaɪə.rᵊn.ses.tər; ˈsɪs.ɪ.tər/ Ⓤⓢ /ˈsaɪ.rᵊn.ses.tɚ/
Lympne	/lɪm/
Woolfardisworthy	/ˈwʊl.zᵊr.i; wʊlˈfɑː.dɪ.swɜː.ði/ Ⓤⓢ /ˈwʊl.zɚ.i; wʊlˈfɑːr.dɪ.swɜː.ði/
Worcester	/ˈwʊs.tər/ Ⓤⓢ /-ɚ/

Welsh place names can be very difficult to decipher for people who do not know the rules that govern the spelling of Welsh. Although Welsh is written using the same alphabet as English, the values of the letters are frequently different, e.g.:

Llanrwst	/hlænˈruːst/
Penmaenmawr	/ˌpen.mənˈmaʊ.ər, -ˈmɔːr/ Ⓤⓢ /-ˈmaʊ.ɚ, -ˈmɔːr/

There is some regularity amongst suffixes in some place names. The suffix *-ham* in British place names, for example, is usually pronounced /-əm/, as in *Birmingham*. However, in the North American place name *Birmingham*, *-ham* is pronounced /-hæm/. Another common suffix in British place names, *-cester*, is usually pronounced /-stər/ Ⓤⓢ /-stɚ/, e.g. as in *Worcester* above (although note also *Cirencester*, in which one possible realization is /-ses.tər/ Ⓤⓢ /-ses.tɚ/). Finally, the suffix *-wick* in e.g. *Warwick* is usually pronounced /-ɪk/ in British place names, although there are exceptions.

nasal consonant

A consonant in which the air escapes only through the nose. For this to happen, two articulatory actions are necessary: firstly, the 'soft palate' (or 'velum') must be lowered to allow air to escape past it and, secondly, a closure must be made in the oral cavity to prevent air from escaping through it. The closure may be at any place of articulation from BILABIAL at the front of the oral cavity to 'uvular' at the back (in the latter case there is contact between the tip of the lowered soft palate and the raised back of the tongue).

Examples for English

English has three commonly found nasal consonants: bilabial, alveolar, and velar, for which the symbols /m n ŋ/ are used. /ŋ/ cannot occur at the beginning of a syllable, e.g.:

map /mæp/
nap /næp/
sang /sæŋ/

There is disagreement over the phonemic status of the velar nasal: some claim that it must be a phoneme since it can be placed in contrastive contexts like *sum/sun/sung*, while others state that the velar nasal is an ALLOPHONE of /n/ which occurs before /k/ and /g/.

In English we find 'nasal release' of PLOSIVE consonants: when a plosive is followed by a nasal consonant the usual articulation is to release the compressed air by lowering the soft palate. This is particularly noticeable when the plosive and the nasal are 'homorganic' (share the same place of articulation), as for example in *topmost*, *Putney*. The result is that no plosive release is heard from the speaker's mouth before the nasal consonant, e.g.:

topmost /ˈtɒp.məʊst/ Ⓤⓢ /ˈtɑːp.moʊst/
Putney /ˈpʌt.ni/

Nasal release can also result in a SYLLABIC CONSONANT, e.g.:

button /ˈbʌtn̩/

nasalization

The addition of nasal escape of air to a sound which would not normally have it.

Examples for English

The best-known examples of nasalization in English are nasalized vowels. In most vowels the airflow escapes entirely through the mouth, but often, in a vowel preceding or following a nasal consonant, we find air escaping also through the nose.

This is a kind of coarticulation, e.g.:

pin /pɪn/ [pʰĩn]
man /mæn/ [mæ̃n]
sing /sɪŋ/ [sĩŋ]

Nasalized vowels are not phonemically contrastive in English.

In other languages

Nasalized vowels are phonemically contrastive in a number of languages, such as French, which has pairs of words such as *beau – bon* /bo – bõ/ and *mais – main* /mɛ – mɛ̃/.

neutral

A term used to describe lip configuration in speech sounds, in which the lips are neither rounded nor

'unrounded' is also commonly used but can apply equally to spread lips.

Examples for English

The English vowels /ə/ and /ɜː/ are thought of as having a neutral lip configuration.

neutralization

The loss of contrast between PHONEMES.

Examples for English

In its simple form, the theory of the phoneme implies that two sounds that are in opposition to each other (e.g. /t/ and /d/ in English) are in this relationship in all contexts throughout the language. Closer study of phonemes has, however, shown that there are some contexts where the opposition no longer functions: for example, in a word like *still* /stɪl/, the /t/ is in a position (following /s/ and preceding a vowel) where voiced (LENIS) PLOSIVES do not occur. There is no possibility in English of the existence of a pair of words such as /stɪl/ and */sdɪl/, so in this context the opposition between /t/ and /d/ is 'neutralized'. One consequence of this is that one could equally well claim that the plosive in this word is a /d/, not a /t/. (See also ASPIRATION.)

onset

In the analysis of syllable structure (and occasionally in other areas), the first part of a syllable.

Examples for English

In English the onset may be zero (when no consonant precedes the vowel in a syllable), one consonant, or two, or three, e.g.:

in /ɪn/
spin /spɪn/
pin /pɪn/
spring /sprɪŋ/

There are many restrictions on what clusters of consonants may occur in onsets: for example, if an English syllable has a three-consonant onset, the first consonant must be /s/ and the last one must be one of /l r j w/.

palatal

A palatal consonant is one in which the tongue makes contact with or approaches the highest part of the hard palate. The hard palate is mainly composed of a thin layer of bone and is dome-shaped, as you can feel by exploring it with the tip of your tongue.

Examples for English

In English, the only phoneme described as palatal is /j/, e.g.:

yes /jes/
beautiful /ˈbjuː.tɪ.fəl/ Ⓤⓢ /-ɚ-/

However, phonetically a voiceless palatal fricative [ç] can also occur for the consonants in the sequence /hjuː/, e.g.:

huge /hjuːdʒ/ [çuːʤ̊]

palato-alveolar

Palato-alveolar sounds are made between the upper teeth and the front part of the palate. In the description of English, this term has been largely replaced by POST-ALVEOLAR.

In other languages

It has been proposed that there is a difference between palato-alveolar and 'alveolo-palatal' that can be reliably distinguished, though others argue that factors other than place of articulation are usually involved. The latter sounds are placed further forward in the mouth than the former: an example of an alveolo-palatal consonant is that of Polish /ɕ/ in *Kasia* (compare /ʃ/ in *kasza*).

pharyngeal

Descriptive of a sound made by constricting the muscles of the pharynx (and usually also some of the LARYNX muscles) to create an obstruction to the airflow from the lungs.

Examples for English

English does not have any pharyngeal phonemes.

In other languages

The best known language that has pharyngeal consonants is Arabic, most dialects of which have voiced and voiceless fricatives.

phone

A unit at the phonetic level in the study of speech. The term PHONEME is very widely used for a contrastive unit of sound in language. However, a term is also needed for a unit at the phonetic level, since there is not always a one-to-one correspondence between units at the two levels.

Examples for English

The English word *can't* is phonemically /kɑːnt/ in British English and /kænt/ in US English (four phonemic units), but may be pronounced [kɑ̃ːt] or [kæ̃t] with the nasal consonant phoneme absorbed into the preceding vowel as NASALIZATION (three phonetic units).

phoneme

A fundamental unit of phonology, usually said to be the smallest unit of speech. It has been defined and used in many different ways.

Examples for English

Virtually all theories of phonology hold that spoken language can be broken down into a string of sound units (phonemes), and that each language has a small, relatively fixed set of these phonemes. Most phonemes can be put into groups; for example, in English we can identify a group of PLOSIVE phonemes /p t k b d g/, a group of voiceless FRICATIVES /f θ s ʃ h/, and so on. An important question in phoneme theory is how the analyst can establish what the phonemes of a language are.

The most widely accepted view is that phonemes are 'contrastive' and one must find cases where the

difference between two words is dependent on the difference between two phonemes. For example, we can prove that the difference between *pin* and *pan* depends on the vowel, and that /ɪ/ and /æ/ are different phonemes.

Pairs of words that differ in just one phoneme are known as 'minimal pairs'. We can establish the same fact about /p/ and /b/ by citing *pin* and *bin*.

Tests like these are called 'commutation tests' and can only be carried out when a provisional list of possible phonemes has been established, so some basic phonetic analysis must precede this stage.

phonetics

The scientific study of speech. It has a long history, going back certainly to well over two thousand years ago. The central concerns in phonetics are the discovery of how speech sounds are produced, how they are used in spoken language, how we can record speech sounds with written symbols and how we hear and recognize different sounds.

In the first of these areas, when we study the production of speech sounds we can observe what speakers do ('articulatory' observation) and we can try to feel what is going on inside our vocal tract ('kinaesthetic' observation). The second area is where phonetics overlaps with PHONOLOGY: usually in phonetics we are only interested in sounds that are used in meaningful speech, and phoneticians are interested in discovering the range and variety of sounds used in this way in all the known languages of the world. This is sometimes known as 'linguistic phonetics'. Thirdly, there has always been a need for agreed conventions for using phonetic symbols that represent speech sounds; the International Phonetic Association has played a very important role in this. Finally, the 'auditory' aspect of speech is very important. The ear is capable of making fine discrimination between different sounds, and sometimes it is not possible to define in articulatory terms precisely what the difference is. A good example of this is in vowel classification. While it is important to know the position and shape of the tongue and lips, it is often very important to have been trained in an agreed set of standard auditory qualities that vowels can be reliably related to (see CARDINAL VOWEL). Another important area is acoustic phonetics which studies the physical properties of speech sounds.

phonology

The study of the sound systems of languages.

The most basic activity in phonology is 'phonemic analysis', in which the objective is to establish what the PHONEMES are and arrive at the 'phonemic inventory' of the language. Very few phonologists have ever believed that this would be an adequate analysis of the sound system of a language: it is necessary to go beyond this. One can look at 'suprasegmental' phonology – the study of STRESS, RHYTHM, and INTONATION. One can go beyond the phoneme and look into the detailed characteristics of each unit in terms of 'distinctive features'. The way in which sounds can combine in a language is studied in 'phonotactics' and in the analysis of syllable structure.

For some phonologists the most important area is the relationships between the different phonemes – how they form groups, the nature of the contrasts between them and how those oppositions may be neutralized (see NEUTRALIZATION). For others, the most important activity is to discover the rules which affect the phonemes of the language and the way they are produced, and to express these rules as economically as possible.

pitch

An auditory sensation which places sounds on a scale from low to high.

When we hear a regularly vibrating sound such as a note played on a musical instrument, or a vowel produced by the human voice, we hear a high pitch if the rate of vibration is high and a low pitch if the rate of vibration is low. Many speech sounds are voiceless (e.g. [s]), and cannot give rise to a sensation of pitch in this way. The pitch sensation that we receive from a voiced sound corresponds quite closely to the frequency of vibration of the vocal folds. However, we usually refer to this vibration frequency as 'fundamental frequency' (which we can measure) in order to distinguish it from the subjective impression of pitch.

Pitch is used in many languages as an essential component of the pronunciation of a word, so that a change of pitch may cause a change in meaning: these are called 'tone languages'. In most languages (whether or not they are tone languages) pitch plays a central role in INTONATION.

plosive

A sound produced by forming a complete obstruction to the flow of air out of the mouth and nose. Normally, this results in a build-up of compressed air inside the chamber formed by the closure. When the closure is released, there is a small explosion that causes a sharp noise.

Examples for English

British English and US English have six plosive consonants, /p t k/ (voiceless) and /b d g/ (voiced).

In syllable-initial position, sounds in the voiceless group /p t k/ are strongly aspirated (see ASPIRATION), and in final position GLOTTALIZATION of these sounds causes a shortening of the preceding vowel. Sounds in the voiced group /b d g/ tend to be 'devoiced' at the beginning and ends of words. At the ends of words, the pairs of sounds /p b/, /t d/, and /k g/ can sound very similar due to this, and one must listen to the length of the vowel to work out which consonant is being produced.

The basic plosive consonant type can be of many different forms: plosives may have any place of articulation, may be voiced or voiceless and may have an 'egressive' (breathing out) or 'ingressive' (breathing in) airflow. The airflow may be from the lungs ('pulmonic'), from the larynx ('glottalic') or generated in the mouth ('velaric'). We find great variation in the release of the plosive.

postalveolar

Descriptive of sounds made between the upper teeth and the front part of the palate.

Examples for English

British and US English have two sets of sounds referred to postalveolar, the fricatives /ʃ ʒ/ and the affricates /tʃ dʒ/. These are also referred to as PALATO-ALVEOLAR, e.g.:

pressure	/ˈpreʃ.ər/	(US) /-ɚ/
pleasure	/ˈpleʒ.ər/	(US) /-ɚ/
church	/tʃɜːtʃ/	(US) /tʃɜːtʃ/
judge	/dʒʌdʒ/	

prefixes

A prefix is an element placed at the beginning of a word to modify or alter its meaning. In general, prefixes do not alter the original pronunciation of the word stem on to which they are affixed, though they may attract secondary stress.

A narrow definition of *prefix* would apply only in the case of words where removal of the prefix would leave a freestanding word (for example, 'unfit' is 'un' + 'fit'), but many treatments of English word stress also treat as prefixes such elements as 'con' in 'contain' or 'in' in 'insert', where 'tain' and 'sert' do not exist independently as words.

Examples

Some examples of words containing unstressed prefixes follow, e.g.:

admire	/ədˈmaɪər/ (US) /-ˈmaɪ.ɚ/
contain	/kənˈteɪn/
desist	/dɪˈsɪst/
undo	/ʌnˈduː/

A prefix may attract secondary stress if it is affixed to a word beginning with an unstressed syllable (e.g. another prefix), e.g.:

undivided	/ˌʌn.dɪˈvaɪ.dɪd/
unforseen	/ˌʌn.fɔːˈsiːn/ (US) /-fɔːrˈ-/

A prefix may be stressed to avoid a clash of two stressed syllables in stress-shift situations, e.g.:

unfair dismissal /ˌʌn.feə dɪˈsmɪs.əl/ (US) /-fer-/

In homographic noun/verb pairs containing prefixes, the prefix is usually stressed in the nominal form and unstressed in the verbal form (see HOMOGRAPHS), e.g.:

insert (*n.*)	/ˈɪn.sɜːt/	(US) /-sɜːt/
(*v.*)	/ɪnˈsɜːt/	(US) /-ˈsɜːt/
record (*n.*)	/ˈrek.ɔːd/	(US) /-ɚd/
(*v.*)	/rɪˈkɔːd/	(US) /-ˈkɔːrd/

retroflex

In a retroflex articulation the tip of the tongue is curled upward and backward.

Examples for English

The /r/ sound of some British and American accents is sometimes described as being retroflex, though in BBC pronunciation the degree of retroflexion is relatively small.

In US English and some accents of southwest England it is common for vowels preceding /r/ (e.g. /uː/ in *moon*, or /ɜː/ in *bird*) to be affected by the consonant so that they have a retroflex quality for most of their duration. This 'r-colouring' is most common in back or central vowels where the forward part of the tongue is relatively free to change shape.

In other languages

Other languages have retroflex consonants with a more noticeable auditory quality, the best known examples being the great majority of the languages of the Indian subcontinent. The sound of retroflex consonants is fairly familiar to English listeners, since first-generation immigrants from India and Pakistan tend to carry the retroflex quality into their pronunciation of English consonants which are alveolar in BBC pronunciation.

rhotic

In rhotic varieties of English pronunciation the /r/ PHONEME is found in all phonological contexts.

Examples for English

In BBC pronunciation, /r/ is only found before vowels, and never before consonants or before a pause (see also LIAISON), e.g.:

red	/red/
around	/əˈraʊnd/
there is	/ðər ɪz/

In US English and other rhotic accents, on the other hand, /r/ may occur before consonants and before a pause, e.g.:

cart	/kɑːt/ (US)	/kɑːrt/
car	/kɑːr/ (US)	/kɑːr/

While the BBC accent is non-rhotic, many accents of the British Isles are rhotic, including most of the south and west of England, much of Wales and all of Scotland and Ireland. Most speakers of American English speak with a rhotic accent, but there are non-rhotic areas including the Boston area, lower-class New York and the Deep South.

rhyme/rime

In the phonological analysis of the syllable, this is a way of referring to the vowel in the middle of the syllable forming its 'peak' plus any sounds following the peak within the syllable (the CODA).

Examples for English

In the word *spoon* the rhyme (or rime) is /uːn/, in *tea* it is /iː/ and in *strengths* it is /eŋθs/ or /eŋkθs/.

Note

The spelling 'rhyme' also refers to a pair of lines that end with the same sequence of sounds in verse. If we examine the sound sequences that must match each other, we find that these consist of the vowel and any final consonants of the last syllable: thus *moon* and *June* rhyme, and the initial consonants of these two words are not important (of course, we do find longer-running rhymes than this in verse, e.g. *ability* rhyming with *senility*).

rhythm

The way events in speech are distributed in time.

Examples for English

Obvious examples of vocal rhythms are chanting as part of games or physical activities. In conversational speech the rhythms are vastly more complicated, but it is clear that the timing of speech is not random. An extreme view (though a quite common one) is that English speech has a rhythm that allows us to divide it up into more or less equal intervals of time called 'feet', each of which begins with a stressed syllable: this is called the 'stress-timed rhythm hypothesis', e.g.:

London	ˈlondon
a return to London	a re | ˈturn to | ˈlondon
a day return to London	a | ˈday re | ˈturn to | ˈlondon

Languages where the length of each syllable remains more or less the same as that of its neighbours whether or not it is stressed are called 'syllable-timed'.

Most evidence from the study of real speech suggests that such rhythms only exist in very careful, controlled speaking, but it appears from psychological research that listeners' brains tend to hear timing regularities even where there is little or no physical regularity.

rounding

A term used to describe lip configuration in speech sounds.

Examples for English

Practically any vowel or consonant may be produced with different amounts of lip rounding. The lips are rounded by muscles that act rather like a drawstring round the neck of a bag, bringing the edges of the lips towards each other. Except in unusual cases, this results not only in the mouth opening adopting a round shape, but also in a protrusion or 'pushing forward' of the lips. In theory, any vowel position (defined in terms of height and frontness/backness) may be produced rounded or unrounded, though we do not necessarily find all possible vowels in natural languages. BBC English has four rounded vowel monophthongs, while US English has three:

pot /pɒt/ Ⓤ /pɑːt/ (unrounded)
put /pʊt/
core /kɔː/ Ⓤ /kɔːr/
coo /kuː/

Consonants, too, may have rounded lips (in [w], the basic consonantal articulation itself consists of lip rounding): this lip rounding in consonants is regarded as a 'secondary articulation', and it is usual to refer to it as 'labialization'. In British English, it is common to find /ʃ/, /tʃ/, and /r/ with lip rounding.

In other languages

Swedish is described as having a rounded vowel without lip protrusion.

schwa

An unstressed central vowel.

Examples for English

One of the most noticeable features of English pronunciation is the phonetic difference between stressed and unstressed syllables. In most languages, any of the vowels of the language can occur in any syllable whether that syllable is stressed or not. In English, however, a syllable which bears no stress is more likely to have one of a small number of weak vowels. The most common weak vowel is one which never occurs in a stressed syllable, the schwa vowel (symbolized ə), which is generally described as being unrounded, central (i.e. between front and back), and mid (i.e. between close and open), e.g.:

appease /əˈpiːz/
syllable /ˈsɪl.ə.bᵊl/
China /ˈtʃaɪ.nə/
mother /ˈmʌð.əʳ/ Ⓤ /ˈmʌð.ɚ/

Statistically, schwa is reported to be the most frequently occurring vowel of English (over 10% of all vowels).

soft palate

The rear part of the roof of the mouth.

Most of the roof of the mouth consists of 'hard palate', which has bone beneath the skin. Towards the back of the mouth, the layer of bone comes to an end but the layer of soft tissue continues for some distance, ending eventually in a loose appendage that can easily be seen by looking in a mirror. This dangling object is the 'uvula', and the layer of soft tissue attached to it is called the soft palate (it is also sometimes known as the 'velum'). In normal breathing the velum is allowed to hang down so that air may pass above it and escape through the nose, but for most speech sounds it is lifted up and pressed against the upper back wall of the throat so that no air can escape through the nose. This is necessary for a PLOSIVE, for example, so that air may be compressed within the vocal tract. However, for NASAL consonants (e.g. [m], [n]) the soft palate must be lowered since air should escape only through the nose in these sounds.

In nasalized vowels (such vowels are found in considerable numbers in French, for example) the soft palate is lowered and air escapes through the mouth and the nose together.

spelling pronunciation

It is well known that English spelling is a poor guide to the pronunciation of a word. In some cases the spelling represents the pronunciation that was current several centuries ago, and in others the spelling has been changed by self-appointed authorities to fit with their views on the origins of a word. Sometimes a 'spelling pronunciation' arises when speakers pronounce a word in a way that is closer to what is suggested by the spelling. One reason for such a change is found where a word which has had a widespread pronunciation that was very different from the spelling becomes less frequently used. When 'waistcoat' referred to a garment that most men wore, it was /ˈweskɪt/; now that waistcoats are much less

often worn, the pronunciation /ˈweɪst.kəʊt/ predominates/. We can see the process in operation in the gradual loss of the traditional pronunciations of many nautical terms as the number of people working in boats diminishes, leaving only amateur sailors to continue the tradition: nowadays 'leeward' is more often /ˈliː.wəd/ than /ˈluː.əd/, 'rowlocks' are /ˈrəʊ.lɒks/ rather than /ˈrɒl.əks/, and you are more likely to hear directions such as 'northeast', 'southwest' pronounced as /nɔːθˈiːst/, /ˈsaʊθˈwest/ than as /nɔːrˈiːst/, /ˈsaʊˈwest/. Not all cases of spelling pronunciation can be explained in this way. The word 'conduit' was pronounced /ˈkɒndɪt/ for centuries, but is now usually /ˈkɒndjʊɪt/, despite the fact that it remains quite a well-known word, and it is hard to see a reason for the change from /t/ to /θ/ in the pronunciation of the name 'Anthony'.

spreading

A term used to describe lip positions in speech, produced by pulling the corners of the mouth away from each other, as in a smile.

Phonetics books tend to be rather inconsistent about this, sometimes implying that any sound that is not rounded has spread lips, but elsewhere treating lip spreading as being different from NEUTRAL lip shape. Vowels with spreading are often referred to as 'unrounded'.

Examples for English

The English vowel /iː/ is thought of as having strong lip spreading, while /ɪ/ has spreading to a lesser degree.

stress

A property of syllables which makes them stand out as more noticeable than others.

Examples for English

Stress is a large topic, which cannot be covered in its entirety here. However, some examples follow.

The position of stress can change the meaning or word class of a word, and so forms part of the phonological composition of the word, e.g.:

import (n) /ˈɪm.pɔːt/ Ⓤ /-pɔːrt/
import (v) /ɪmˈpɔːt/ Ⓤ /-ˈpɔːrt/
record (n) /ˈrek.ɔːd/ Ⓤ /-ərd/
record (v) /rɪˈkɔːd/ Ⓤ /-ˈkɔːrd/

It is necessary to consider what factors make a syllable count as stressed. It seems likely that stressed syllables are produced with greater effort than unstressed, and that this effort is manifested in the air pressure generated in the lungs for producing the syllable and also in the articulatory movements in the vocal tract. These effects of stress produce in turn various audible results: one is 'pitch prominence', in which the stressed syllable stands out from its context (for example, being higher if its unstressed neighbours are low in PITCH, or lower if those neighbours are high; often a pitch glide such as a fall or rise is used to give greater pitch prominence). Another effect of stress is that stressed syllables tend to be longer – this is very noticeable in English, less so in some other languages. Also, stressed syllables tend to be louder than unstressed, though experiments have shown that differences in loudness alone are not very noticeable to most listeners. It has been suggested by many writers that the term 'accent' should be used to refer to some of the manifestations of stress (particularly pitch prominence), but the word, though widely used, has never acquired a distinct meaning of its own.

One of the areas in which there is little agreement is that of 'levels' of stress. Some descriptions of languages manage with just two levels (stressed and unstressed), while others use more. In English, one can argue that if one takes the word *indicator* as an example, the first syllable is the most strongly stressed, the third syllable is the next most strongly stressed and the second and fourth syllables are weakly stressed, or unstressed. This gives us three levels: it is possible to argue for more, though this rarely seems to give any practical benefit.

stress shift

A change in the position of the stress in a word when that word is combined with others in a phrase.

Examples for English

The RHYTHM of English prefers patterns in which two stressed syllables do not come together. In order to avoid this, stress in some polysyllabic words may move to an earlier syllable when combined with another in a phrase, e.g.:

Heathrow /hiːθ.rəʊ/ Ⓤ /-roʊ/
Heathrow Airport /ˌhiːθ.rəʊ eə.pɔːt/ Ⓤ /-roʊ er.pɔːrt/
academic /ˌæk.ə.dem.ɪk/
academic dress /ˌæk.ə.dem.ɪk ˈdres/

In this dictionary, words which change their stress in this way are shown with an example demonstrating the stress shift.

suffixes

A suffix is an element placed at the end of a word to modify or alter its meaning. Unlike PREFIXES, it is possible for suffixes to alter the original pronunciation of the word stem on to which they are affixed. This depends on whether the suffix is stress-neutral, pre-stressed or stress-attracting.

Examples

Some suffixes do not change the pronunciation of the word stem. These are known as 'stress-neutral' suffixes. Some words containing stress-neutral suffixes follow:

eleventh /ɪˈlev.ənɪθ/
dramatize /ˈdræm.ə.taɪz/ Ⓤ /ˈdrɑː.mə-/
fatherhood /ˈfɑː.ðə.hʊd/ Ⓤ /-ðɚ-/
happily /ˈhæp.ɪ.li/

A suffix which attracts stress is known as 'stress-attracting'.

Some words containing stress-attracting suffixes follow:

engineer /ˌen.dʒɪˈnɪəʳ/ Ⓤ /-ˈnɪr/
Japanese /ˌdʒæp.əˈniːz/
nineteen /ˌnaɪnˈtiːn/

A 'pre-stressed' suffix is one in which the affixation of the suffix causes stress to be assigned to a syllable before it. There are a number of different types of pre-stressed suffixes. Here is an example where the stress falls on the syllable immediately before the suffix:

despotic /dɪˈspɒt.ɪk/ (US) /desˈpɑː.ṭɪk/

In other words, the stress falls two syllables ahead of the suffix:

insecticide /ɪnˈsek.tɪ.saɪd/

There are also 'mixed' pre-stressed suffixes where the stress may fall either one or two syllables before the suffix.

The suffix *-ation* is actually a combination of the stress-neutral *-ate* and the pre-stressed suffix *-ion*. In words containing *-ation*, the strongest stress is always on the penultimate syllable, e.g.:

condemnation /ˌkɒn.demˈneɪ.ʃᵊn/ (US) /ˌkɑːn-/

syllabic consonant

A consonant which can stand alone as a syllable.

Examples for English

The great majority of syllables in all languages have a vowel at their centre, and may have one or more consonants preceding and following the vowel (though languages differ greatly in the possible occurrences of consonants in syllables). However, in a few cases we find syllables which contain nothing that could conventionally be classed as a vowel. In English, syllabic consonants appear to arise as a consequence of a weak vowel becoming lost, and some appear to have become obligatory in present-day speech, e.g.:

bottle /ˈbɒt.l̩/ (US) /ˈbɑː.ṭl̩/

In many other cases in English it appears to be possible either to pronounce /m n ŋ l r/ as syllabic consonants or to pronounce them with a preceding vowel, e.g.:

button /ˈbʌt.n̩/
orderly /ˈɔː.dl̩.i, -də.li/ (US) /ˈɔːr.də.li/
history /ˈhɪs.tr̩.i, -tə.ri/ (US) /-tr̩.i, -tɚ-/

In this dictionary, the use of a superscript schwa (ᵊ) indicates the possibility of a syllabic consonant.

The matter is more confusing because of the fact that speakers do not agree in their intuitions about whether a consonant (particularly /l/) is syllabic or not: while most would agree that, for example, *cuddle* and *cycle* are disyllabic (i.e. contain two syllables), *cuddly* and *cycling* are disyllabic for some people (and therefore do not contain a syllabic consonant) while for others they are trisyllabic.

In other languages

For syllables not to contain a vowel is a normal state of affairs in some languages (consider the first syllables of the Czech names *Brno* and *Vltava*). In Japanese some consonants appear to be able to stand as syllables by themselves, according to the intuitions of native speakers who are asked to divide speech up into rhythmical beats.

syllable

A fundamentally important unit – the most basic unit in speech. Here we are concerned with the phonological notion of the syllable.

Examples for English

Phonologists are interested in the structure of the syllable, since there appear to be interesting observations to be made about which phonemes may occur at the beginning, in the middle, and at the end of syllables. In English, it is possible to have from zero to up to three consonants in the ONSET of a syllable, and from zero to up to four in the CODA.

The study of sequences of phonemes is called 'phonotactics', and it seems that the phonotactic possibilities of a language are determined by syllabic structure. This means that any sequence of sounds that a native speaker produces can be broken down into syllables without any segments being left over. For example, in *Their strengths triumphed frequently*, we find the rather daunting sequences of consonant phonemes /ŋθstr/ and /mftfr/, but using what we know of English phonotactics we can split these clusters into one part that belongs to the end of one syllable and another part that belongs to the beginning of another. Thus the first one can only be divided /ŋθ | str/ or /ŋθs | tr/ and the second can only be /mft | fr/. Phonological treatments of syllable structure usually call the first part of a syllable the ONSET, the middle part the 'peak' and the end part the CODA. The combination of peak and coda is called the RHYME. Syllable breaks, however, may be problematic when approximants occur at syllable boundaries.

tap

A sound which resembles [t] or [d], being made by a complete closure between the tongue and the alveolar region, but which is very brief and is produced by a sharp upward throw of the tongue blade. As soon as contact is made, the effects of gravity and air pressure cause the tongue to fall again.

Examples for English

The tap sound (for which the phonetic symbol is [ɾ]) is noticeable in Scottish accents as the realization of /r/, and in US English it is often heard as a (voiced) realization of /t/ when it occurs after a stressed vowel and before an unstressed one, e.g.:

getting better (US) /ˈgeṭ.ɪŋ ˈbeṭ.ɚ/ [ɡ̞ɛɾɪŋb̞ɛɾɚ]

In British English it used to be quite common to hear a tap for /r/ in careful or emphatic speech (e.g. *very* [ɣeɾɪ]), though this is less often heard now. It is increasingly common to hear the American-style tapped /t/ in England.

Several varieties of tap are possible: they may be voiced or voiceless. For instance, Scottish pre-pausal /r/ is often realized as a voiceless tap, as in *here* [hiɾ̥ᵊ]. Taps may also be produced with the SOFT PALATE lowered, resulting in a nasalized tap which is sometimes heard in the US English pronunciation of words like *mental* [mẽɾəɫ]. A closely related sound is the FLAP, and the TRILL also has some similar characteristics.

tense

See LAX.

tone

An identifiable movement or level of PITCH that is used in a linguistically contrastive way.

Examples for English

In English, tone forms the central part of INTONATION, and the difference between, for example, a rising and a falling tone on a particular word may cause a different interpretation of the sentence in which it occurs, e.g.:

- it was ˋcold
  (meaning 'it was cold')
- it was ˇcold
  (meaning 'was it cold?')

There are often recognized as being at least five tones in English: a fall, a rise, a fall-rise, a rise-fall, and a level tone.

Meanings are frequently ascribed to each tone; the scope of this glossary does not allow us to discuss this further. In intonation, tone may be spread over many syllables, e.g., in:

- his ˇcar could have broken down
  (in which the pitch movement falls on *car* and rises on *down*)

In other languages

In some languages (known as 'tone languages') the linguistic function of tone is to change the meaning of a word: in Mandarin Chinese, for example, [ma] said on a high pitch means *mother* while [ma] said on a low rising tone means *hemp*. It is usual to identify tones as being a property of individual syllables.

tone unit

A unit of speech consisting of one or more syllables or feet.

Examples for English

A tone unit must contain a tonic syllable, that is, a syllable on which a pitch movement begins. Only one tonic syllable is allowed in an English tone unit. The tone unit may also contain a 'head' (from the first stressed syllable up to the tonic syllable), a 'pre-head' (any unstressed syllables preceding the head), and a 'tail' (all syllables after the tonic syllable). The tonic syllable is underlined in each of the examples which follow:

- ˋyes
  tonic
- ˈJoe said ˋyes
  head + tonic
- and then ˈJoe said ˋyes
  pre-head + head + tonic
- and then ˈJoe said ˋyes to me
  pre-head + head + tonic + tail

As each of the non-tonic elements is optional, it is possible to have any combination together with the tonic syllable, e.g.:

- was it ˋyou
  pre-head + tonic
- it was ˋyesterday
  pre-head + tonic + tail
- ˈnow I understand
  tonic + tail

In the study of INTONATION it is usual to divide speech into larger units than syllables. If one studies only short sentences said in isolation it may be sufficient to make no subdivision of the utterance, but in longer utterances there must be some points at which the analyst marks a break between the end of one pattern and the beginning of the next. These breaks divide speech into 'tone units', and are called 'tone-unit boundaries', e.g.:

| the ˋlast time I saw her | was ˋyesterday

If the study of intonation is part of phonology, these boundaries should be identifiable with reference to their effect on pronunciation rather than to grammatical information about word and clause boundaries; statistically, however, we find that in most cases tone-unit boundaries do fall at obvious syntactic boundaries, and it would be rather odd to divide two tone units in the middle of a phrase. The most obvious factor to look for in trying to establish boundaries is the presence of a pause, and in slow careful speech (e.g. in lectures, sermons, and political speeches) this may be done quite regularly. However, it seems that we detect tone-unit boundaries even when the speaker does not make a pause, if there is an identifiable break or discontinuity in the rhythm or in the intonation pattern.

There is evidence that we use a larger number of shorter tone units in informal conversational speech, and fewer, longer tone units in formal styles.

trill

A speech sound produced by the rapid vibration of one of the vocal organs.

Examples

The parts of the body that are used in speaking (the 'vocal apparatus') include some 'wobbly bits' that can be made to vibrate. When this type of vibration is made as a speech sound, it is called a trill. The possibilities include a BILABIAL trill, where the lips vibrate (used as a mild insult, this is sometimes called 'blowing a raspberry', or, in the US, a 'Bronx cheer'), a tongue-tip trill which is produced in many languages for a sound represented alphabetically as r, and a uvular trill, which is a rather dramatic way of pronouncing a 'uvular r' as found in French, German, and many other European languages, most commonly used in acting and singing.

In British English, the trill most likely to occur is the ALVEOLAR trill, which is (perhaps confusingly) represented by the symbol [r], and is an allophone of the English phoneme /r/. However, it most frequently occurs in restricted contexts, such as singing.

triphthong

A vowel glide within a word that has three distinguishable vowel qualities.

Examples for English

In British English there are five triphthongs, formed by combining the diphthongs /eɪ aɪ ɔɪ aʊ əʊ/ with /ə/. It is not always easy to decide if they form one syllable or two. The following examples are given without syllable divisions:

layer	/leɪə'/ (US) /'leɪɚ/	
liar	/laɪə'/ (US) /'laɪɚ/	
loyal	/lɔɪəl/ (US) /'lɔɪəl/	
power	/paʊə'/ (US) /paʊɚ/	
mower	/məʊə'/ (US) /'moʊɚ/	

Other combinations of three vowel qualities, such as /eɪʊ/ in *play-off* /'pleɪʊf/ ((US) /eɪɑː/ in /'pleɪɑːf/), or /iːəʊ/ in *reopen* /ˌriː'əʊpən/ ((US) /iːoʊ/ in *reopen* /ˌriː'oʊpən/) do not seem to form such tightly knit units, though from the strictly phonetic point of view they are triphthongs.

Syllable division

For General American it can be assumed that these triphthongs have two syllables, and the above examples can be written with syllable divisions: /'leɪ.ɚ/, /'laɪ.ɚ/, /'lɔɪ.əl/, /paʊ.ɚ/, /'moʊ.ɚ/. In British English, however, it is not always possible to decide whether a triphthong is monosyllabic or disyllabic. In some northern accents the division into two syllables is so clear that a transcription with /j/ or /w/ would be suitable (e.g. 'fire' /'fa.jə'/, 'power' /'pa.wə'/, but in BBC pronunciation some words seem to be monosyllabic, though not all speakers. Thus 'fire' is often a single syllable /faɪə'/ with a very weak /ɪ/ in the middle, though some speakers pronounce it as two syllables /'faɪ.ə'/. Where both possible pronunciations exist, they are both shown in this dictionary.

uvular

A consonant sound made between the back of the tongue and a lump of soft tissue (uvula) which is in the back of the mouth, dangling from the end of the soft palate.

Examples for English

English does not have any examples of uvular sounds as phonemes, but sounds with this place of articulation are widely found in other languages.

In other languages

The voiceless uvular PLOSIVE [q] is found as the phoneme /q/ in many dialects of Arabic. Uvular FRICATIVES [χ ʁ] are found quite commonly: German, Hebrew, Dutch, and Spanish, for example, have voiceless ones, and French, Arabic, and Danish have voiced ones. The uvular NASAL [ɴ] is found in some Inuit languages. The uvula itself is active only when it vibrates in a uvular TRILL, [ʀ].

velar

Velar consonant sounds are produced between the tongue and the SOFT PALATE, or 'velum'.

Examples for English

Velar sounds occurring as English phonemes are /k/, /g/, and /ŋ/. The first two are PLOSIVES, and the last, a NASAL. Although /k/ and /g/ can occur syllable-initially and finally, /ŋ/ is restricted to syllable-final position only in English, e.g.:

cap	/kæp/
gap	/gæp/
sack	/sæk/
sag	/sæg/
sang	/sæŋ/

/k/ and /g/ are one of the pairs of consonants said to be distinguished from each other by being FORTIS or LENIS rather than voiced or voiceless.

velarization

A type of secondary articulation in which a constriction in the vocal tract is added to the primary constriction which gives a consonant its place of articulation.

Examples for English

In the case of English DARK L, the /l/ phoneme is articulated with its usual primary constriction in the alveolar region, while the back of the tongue is raised as for an [u] vowel creating a secondary constriction (see DARK L).

In other languages

Arabic has a number of consonant phonemes that are velarized, and are known as 'emphatic' consonants.

vocal folds

An essential part of the larynx, performing a number of important linguistic functions. The term 'vocal cords' is also used.

The vocal folds may be firmly closed to produce what is sometimes called a GLOTTAL STOP, and while they are closed the larynx may be moved up or down to move air out of or into the vocal tract; this is done in the production of 'ejective' and 'implosive' consonants. When brought into light contact with each other the vocal folds tend to vibrate if air is forced through them, producing VOICING, also called 'phonation'. This vibration can be made to vary in many ways, resulting in differences in such things as PITCH, loudness, and voice quality. If a narrow opening is made between the vocal folds, friction noise can result, and this is found in whispering and in the glottal fricative [h]. A more widely open glottis is found in most voiceless consonants.

voice onset time

A measure of the timing of the start of voicing.

All languages distinguish between voiced and voiceless consonants, and PLOSIVES are the most common consonants to be distinguished in this way. However, this is not a simple matter of a plosive being either completely voiced or completely voiceless: the timing of the voicing in relation to the consonant articulation is very important. In one particular case this is so noticeable that it has for a long time been given its

voicing does not happen until some time after the release of the plosive (usually voiceless). This delay, or 'lag', has been the subject of much experimental investigation which has led to the development of a scientific measure of voice timing called voice onset time or V.O.T. The onset of voicing in a plosive may lag behind the plosive release, or it may precede ('lead') it, resulting in a fully or partially voiced plosive. Both can be represented on the V.O.T. scale, one case having positive values and the other negative values.

voicing

A term used to refer to the vibration of the VOCAL FOLDS.

A sound made with this vibration is called *voiced*, and one without vibration is called *voiceless*. Most vowels in most languages are voiced, though voiceless vowels are found in some languages. Among consonants, it is very common to find pairs that are distinguished from each other by the presence or absence of voicing, but in English we find that consonants such as /b d g v ð z ʒ/, though frequently described as voiced (in contrast with the corresponding voiceless consonants /p t k f θ s ʃ/) often have little or no voicing. See FORTIS, LENIS, VOICE ONSET TIME. Whispering is speech that is entirely voiceless.

In other languages

Many pairs of consonants distinguished by the voiced/voiceless contrast are found in different languages. Welsh has voiced and voiceless LATERAL consonants, while Burmese has voiced and voiceless NASAL consonants.

vowel

The class of sound which makes the least obstruction to the flow of air. They are almost always found at the centre of a syllable, and it is rare to find any sound other than a vowel which is able to stand alone as a whole syllable.

Examples for English

Here we examine the vowel monophthongs of English.

In phonetic terms, each vowel has a number of properties that distinguish it from other vowels. These include the shape of the lips, which may be rounded (as for an [u] vowel), NEUTRAL (as for [ə]) or spread (as in a smile, or an [i] vowel – photographers traditionally ask their subjects to say *cheese* /tʃiːz/ so that they will seem to be smiling):

Unrounded	Rounded	(US) Unrounded	Rounded
iː	uː	iː	uː
ɪ	ʊ	ɪ	ʊ
e	ɔː	e	ɔː
æ	ɒ	æ	
ʌ		ʌ	
ɑː		ɑː	
ɜː		ɜː	
ə		ɚ ɝ	

Secondly, the front, the middle, or the back of the tongue may be raised, giving different vowel qualities: the BBC English /æ/ vowel (*cat*) is a front vowel, while the /ɑː/ of *cart* is a back vowel:

Front	Central	Back	(US) Front	Central	Back
iː	ʌ	uː	iː	ʌ	uː
ɪ	ɜː	ʊ	ɪ	ɜː	ʊ
e	ə	ɔː	e	ə ɚ	ɔː
æ		ɒ	æ		ɑː
		ɑː			

The tongue (and the lower jaw) may be raised close to the roof of the mouth, or the tongue may be left low in the mouth with the jaw comparatively open. In British phonetics we talk about 'close' and 'open' vowels, whereas American phoneticians more often talk about 'high' and 'low' vowels. The meaning is clear in either case, e.g.:

Close	Mid	Open	(US) High	Mid	Low
iː	e	ʌ	iː	e	ʌ
ɪ	ɜː	ɒ	ɪ	ɜː	ɑː
uː	ə	ɑː	uː	ə ɚ	æ
ʊ	ɔː	æ	ʊ	ɔː	

Vowels also differ in other ways: they may be 'nasalized' (see NASALIZATION) by being pronounced with the soft palate lowered as for [m], [n], or [ŋ]. Vowels may be voiced, as the great majority are, or voiceless: in English the first vowel in *perhaps* or *potato* is often voiceless. It is claimed that in some languages (including English) there is a distinction to be made between TENSE and LAX vowels, the former being made with greater force than the latter.

In other languages

Nasalization is phonemically contrastive in French, where we find 'minimal pairs' such as *très* /tʀe/ (*very*) and *train* /tʀɛ̃/ (*train*), where the [̃] diacritic indicates nasality. Concerning voiceless vowels: in Portuguese, for example, unstressed vowels in the last syllable of a word are often voiceless. Less usual is the case of stressed voiceless vowels, but these are found in French: close vowels, particularly /i/ but also the close front rounded /y/, become voiceless for some speakers when they are word-final before a pause (for example *oui* [wi], *midi* [midi̥], and also *entendu* [ɑ̃tɑ̃dy̥]).

vowel reduction

The process by which an unstressed vowel may change to become like the mid central vowel 'schwa' /ə/. In the words 'photograph', 'photographic', and 'photography', the vowels in the first three syllables alternate between full vowels and /ə/ according to the position of stress, e.g.:

photograph	/'fəʊ.tə.ɡrɑːf/
photographic	/ˌfəʊ.tə'ɡræf.ɪk/
photography	/fə'tɒɡ.rə.fi/

Although the word 'man' is pronounced /mæn/, the reduced form can be seen in 'postman' /'pəʊst.mən/, 'chairman' /'tʃeə.mən/, etc.

Another example of vowel reduction is found in the alternation between /e/ and /ɪ/ in, for example, the first syllables of 'exhibition' /ˌek.sɪ'bɪʃ.ən/ and /ɪɡ'zɪb.ɪt/.

Symbols used for the transcriptions

Vowel phonemes:

British	American
ɪ as in 'kit'	ɪ as in 'kit'
e as in 'dress'	e as in 'dress'
æ as in 'trap'	æ as in 'trap'
ʌ as in 'strut'	ʌ as in 'strut'
ʊ as in 'foot'	ʊ as in 'foot'
ɒ as in 'lot'	
ə as in 'potato', 'America'	ə as in 'potato', 'America'
	ɚ as in 'survivor'
iː as in 'fleece'	iː as in 'fleece'
ɑː as in 'father'	ɑː as in 'father'
ɔː as in 'thought'	ɔː as in 'thought'
uː as in 'goose'	uː as in 'goose'
ɜː as in 'nurse'	ɜ˞: as in 'nurse'
eɪ as in 'face'	eɪ as in 'face'
aɪ as in 'price'	aɪ as in 'price'
ɔɪ as in 'choice'	ɔɪ as in 'choice'
əʊ as in 'goat'	oʊ as in 'goat'
aʊ as in 'mouth'	aʊ as in 'mouth'
ɪə as in 'near'	
eə as in 'square'	
ʊə as in 'cure'	

Consonant phonemes (British and American)

p as in 'pea'	b as in 'bee'
t as in 'tea'	d as in 'do'
k as in 'key'	g as in 'go'
f as in 'fat'	v as in 'vat'
θ as in 'thin'	ð as in 'that'
s as in 'sip'	z as in 'zip'
ʃ as in 'ship'	ʒ as in 'measure'
h as in 'hat'	l as in 'led'
m as in 'map'	r as in 'red'
n as in 'nap'	w as in 'wet'
ŋ as in 'hang'	j as in 'yet'
tʃ as in 'chin'	dʒ as in 'gin'

Non-phonemic symbols for English

i as in 'react', 'happy'

u as in 'to each', 'influenza'

ᵊ (superscript schwa) as in 'sudden' /ˈsʌd.ᵊn/: vowel may be pronounced, or may be omitted with the following consonant becoming syllabic

ʳ (superscript r): at the end of a word, potential 'linking r', e.g. 'car' /kɑːʳ/ (British English)

l̩ ('syllabic l') as in 'muddle' /ˈmʌd.l̩/

ɾ ('tapped t'), American pronunciation, as in 'better' /ˈbeɾ.ɚ/

Symbols for other languages

x (voiceless velar fricative) as in 'loch', 'Bach'

æ̃, ɑ̃ː, ɔ̃ː, ɜ̃ː (nasalized vowels) as in French 'vin', 'restaurant', 'bon', 'Lebrun'

Prosodic marks

ˈ (primary stress) as in 'very' /ˈver.i/

ˌ (secondary stress) as in 'repaint' /ˌriːˈpeɪnt/

. (syllable division) as in 'happy' /ˈhæp.i/

Italic symbols

Italic symbols are optional sounds, which may be pronounced or omitted, as in 'lunch' /lʌntʃ/